frontiers in Psychiatry

TYPE Original Research
PUBLISHED 29 September 2026
DOI 10.3389/fpsyt.2026.1940193

Check for updates

OPEN ACCESS

EDITED BY
Toshiyuki Ohtani,
Chiba University, Japan

REVIEWED BY
Joanna S. Cohen,
Johns Hopkins University, United States
Laura Hull,
University College London,
United Kingdom

*CORRESPONDENCE
Monika Baker
monika.baker@hsc.utah.edu

RECEIVED 17 July 2026
REVISED 24 August 2026
ACCEPTED 26 August 2026
PUBLISHED 29 September 2026

CITATION
Baker M, Lam W-y, Virtosu M and de Lacy N (2026) Suicide attempt risk in autism: a national EHR study of 2.3 million individuals.
Front. Psychiatry 17:1940193.
doi: 10.3389/fpsyt.2026.1940193

COPYRIGHT
© 2026 Baker, Lam, Virtosu and de Lacy. This is an open-access article distributed under the terms of the Creative Commons Attribution License (CC BY). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

Suicide attempt risk in autism: a national EHR study of 2.3 million individuals

Monika Baker[1]*, Wai-yin Lam[1], Mihai Virtosu[2] and Nina de Lacy[1,3]

[1]Department of Psychiatry, University of Utah, Salt Lake City, UT, United States, [2]Data Science Services, University of Utah, Salt Lake City, UT, United States, [3]Huntsman Mental Health Institute, Salt Lake City, UT, United States

Background: Suicide attempts (SA) are elevated in individuals with autism spectrum disorder (ASD), but population-level data characterizing how SA prevalence varies across demographic and clinical subgroups — at the scale and granularity needed to inform evidence-based risk stratification — have been largely unavailable. This study examines SA prevalence across sex, age group, psychiatric comorbidity type, and substance use disorder subtype in the largest real-world ASD cohort to date.

Methods: We conducted a retrospective cross-sectional analysis using Epic Cosmos electronic health record data from 2,311,171 individuals with ASD identified by International Classification of Diseases, Tenth Revision, Clinical Modification (ICD-10-CM) codes, spanning 2016–2025. SA prevalence was calculated with Wilson score 95% confidence intervals. Modified Poisson regression with robust variance estimation was used to estimate adjusted prevalence ratios (aPRs) for sex, age group, and psychiatric comorbidity. Unadjusted prevalence ratios (uPRs) were calculated separately for individual comorbidity types and substance use disorder subtypes. Supplementary sensitivity analyses were conducted to examine SA prevalence by intellectual disability (ID) status and race/ethnicity.

Results: Overall SA prevalence was 1.7% (38,160 individuals). Females showed higher SA prevalence than males (2.9% vs. 1.2%; aPR 1.61, 95% CI 1.35–1.92). SA prevalence peaked in the 15–24 age group overall (aPR 5.14, 95% CI 3.62–7.29), with sex-stratified analyses revealing that females peaked earlier (15–24 years; aPR 4.33, 95% CI 4.23–4.43) than males (25–34 years; aPR 6.08, 95% CI 5.05–7.31) — a sex-specific divergence in the timing of peak SA prevalence not previously documented in ASD. Having at least one psychiatric comorbidity was associated with a 30-fold higher SA prevalence (aPR 30.56, 95% CI 26.03–35.89), with the effect stronger in females (aPR 36.54, 95% CI 31.04–43.01) than males (aPR 27.77, 95% CI 22.65–34.06). Among comorbidity categories, substance-related disorders showed the highest crude SA prevalence (16.9%; uPR 134.20, 95% CI 67.68–266.10). Substance use disorder subtype analysis revealed SA prevalence ranging from 19.7% to 24.3% across all five substance use disorder subtypes examined, with stimulant use disorder showing the highest unadjusted prevalence ratio of any subtype (uPR 196.11, 95% CI 100.63–382.20). In supplementary sensitivity analyses, SA prevalence was higher among individuals with ID (2.37–2.38%) than without (1.58%) and ranged by race/ethnicity from 0.65–0.92% (Non-Hispanic Other Race) to 2.25–2.28% (Non-Hispanic Multi-race).

Conclusion: SA prevalence in ASD is markedly elevated relative to the general population and varies meaningfully by sex, age, comorbidity profile, intellectual disability status, and race/ethnicity in clinically important ways. Females carry a disproportionate SA burden relative to males, with peak vulnerability arriving earlier in adolescence; males peak later in young adulthood and remain at elevated risk into midlife. Psychiatric comorbidity — particularly substance use disorders — is associated with the largest relative elevations in SA prevalence. In supplementary analyses, individuals with intellectual disability showed elevated SA prevalence relative to those without, and prevalence varied substantially by race/ethnicity, with the highest burden observed among Non-Hispanic Multi-race individuals. These population-level estimates are directly applicable to EHR-based risk stratification models and can inform the development of ASD-specific clinical decision support tools that concentrate surveillance and intervention on those at demonstrably elevated risk, rather than applying uniform approaches across a heterogeneous population.

KEYWORDS

autism spectrum disorder, electronic health records, risk stratification, substance use disorder, suicide attempt

1 Introduction

Autism Spectrum Disorder (ASD) is a continuum of complex neurodevelopmental disorders influenced by various environmental and genetic factors, with an etiology that remains incompletely understood (1). While ASD has significant phenotypic heterogeneity (2), it is characterized by differences in social communication and interaction, alongside restricted and repetitive patterns of behavior (3). ASD is associated with multiple comorbidities including attention deficit hyperactivity disorder (ADHD), mood disorders, Obsessive-Compulsive Disorder (OCD), and schizophrenia (4, 5). Beyond these comorbid conditions, persons with ASD face elevated rates of suicide attempts (SA) relative to the general population (6). A 2024 meta-analysis estimated that individuals with ASD face a 2.85-fold higher risk of death by suicide compared to the general population (7). Pooled prevalence estimates from recent meta-analyses indicate that between 15–24% of individuals with ASD report a lifetime history of SA (6, 8, 9). The elevated SA burden in ASD is thought to reflect a confluence of ASD-specific vulnerabilities, including chronic social isolation, discrepancies in how distress is communicated to persons without ASD, and the psychological costs of masking autistic traits (10, 11).

Beyond overall prevalence, the demographic and clinical correlates of SA in ASD — including variation by sex, age, and psychiatric comorbidity — remain incompletely characterized (6, 9), in part because much of the existing literature conflates suicide attempts with suicidal ideation, self-harm, or suicidal behaviors broadly (6, 9, 12–14) — outcomes that are conceptually and clinically distinct, with evidence that their associated factors differ meaningfully (15, 16). Studies that aggregate these outcomes risk obscuring the specific correlates of SA and limiting the clinical utility of findings for identifying individuals at greatest risk (17). This is especially evident in the literature on sex differences in SA risk, where findings vary considerably depending on which outcome is examined (18, 19). Males and females with ASD show similar risk of death by suicide (18), while more recent large population-based studies have identified higher SA risk in females (8, 18, 20), a pattern that has only emerged as research in this area has grown (19). Autism has historically been characterized as a predominantly male condition, with diagnosis rates and much of the foundational research skewed toward males (18, 21). As diagnostic recognition of ASD in females has improved over the past two decades, with female diagnosis rates increasing steadily and females still consistently diagnosed later than males (21), prior estimates of sex differences in SA risk are likely underestimated in older studies conducted when female ASD diagnoses were less common (19, 21).

Age is an important demographic correlate of SA risk in ASD, with adolescence and young adulthood consistently identified as a period of heightened vulnerability reflecting the confluence of increased social demands, diagnostic transitions, and reduced structured support during this developmental period (22, 23). However, the ASD suicide literature has been heavily skewed toward youth, with research on middle-aged and older adults with ASD accounting for only 0.4% of indexed autism research — a significant gap given emerging evidence that suicidality remains elevated and may even increase with age in individuals with ASD (24–26). Population-level, age-stratified SA prevalence data across the full lifespan therefore remain limited, and whether age patterns of risk differ by sex in ASD has not been characterized at scale (6, 9).

Psychiatric comorbidity is among the most consistently identified correlates of SA in ASD, with any co-occurring psychiatric disorder associated with elevated SA risk across multiple population-based studies and systematic reviews (9, 12, 20). Whether this association is disproportionately stronger in ASD than in the general population is unclear. General-population estimates place the increase in suicide risk associated with any psychiatric disorder at 13-fold (27). In a Finnish national birth cohort of 4,695 autistic individuals matched to 18,450 non-autistic controls, the elevated risk of intentional self-harm in autistic individuals was no longer

statistically significant after adjusting for co-occurring psychiatric disorders (adjusted HR 0.8, 95% CI 0.4–1.6), and comorbidity increased self-harm risk at similar magnitudes in both autistic and non-autistic groups across most categories examined (28). Mood disorders represent the most extensively studied comorbidity in this context — depressive symptoms are among the most frequently and consistently identified correlates of suicidality in ASD, though existing evidence largely pertains to suicidal ideation rather than suicide attempts specifically, reflecting the broader measurement heterogeneity in this literature (9, 22, 29). Co-occurring psychotic disorders have similarly been associated with markedly elevated SA risk in ASD, a relationship thought to reflect compounding vulnerability from both conditions given the well-established link between psychosis and suicidality in the general population (20, 30). Trauma and stressor-related disorders warrant particular attention in the ASD context. Individuals with ASD experience bullying victimization at roughly three times the rate of peers without ASD (70) and face elevated exposure to victimization and adverse childhood events more broadly (11). PTSD has been identified as one of the most strongly associated comorbidities with SA in ASD specifically (11, 20). Eating disorders similarly confer markedly elevated SA risk in the general population — with individuals with anorexia nervosa 18 times more likely to die by suicide relative to age- and sex-matched peers — yet their specific association with SA in ASD has received limited empirical attention despite meaningful phenotypic overlap between ASD and eating disorders (12, 31). Substance use disorders have been identified as among the strongest comorbidity-associated correlates of SA in ASD (20), yet existing studies have largely examined substance use disorders as a broad category without characterizing risk across specific subtypes such as stimulant, cannabis, opioid, and alcohol use disorders — limiting the granularity of findings available to inform clinical decision-making (12, 20).

Race, ethnicity, and intellectual disability (ID) status each show documented, though complex, relationships with suicide-related outcomes. In the general population, racial and ethnic differences in suicidal ideation and SA do not follow a uniform pattern: Black and Hispanic adolescents have been found to report lower rates of suicidal ideation than White adolescents, but higher rates of SA among those endorsing ideation, suggesting differential pathways from ideation to attempt across racial/ethnic groups (32). Among adults with ASD specifically, Black individuals have been found to experience distinct depression symptom burden and disparities in mental health service use relative to White individuals, raising the possibility that similar disparities may extend to SA risk, though this has not been directly established in ASD populations (33). The relationship between co-occurring ID and suicidality in ASD is complex, depending on both outcome measured and ID severity. In Medicare-enrolled autistic adults, co-occurring ID (any level) was associated with lower odds of suicidal ideation but higher odds of suicide attempt (34). Among autistic youth in emergency departments, ID as a collapsed category showed no significant association with a suicidal ideation or self-inflicted injury diagnosis, but severity-specific analysis revealed a non-monotonic pattern: mild ID was associated with significantly higher odds relative to no co-occurring ID, while severe and unspecified ID were associated with significantly lower odds (35).

The burden of SA in ASD has important clinical implications that remain incompletely addressed, as evidence-based prevention strategies and validated assessment approaches specific to ASD are scarce (9, 11). Universal suicide risk screening — the application of standardized screening tools to all individuals in a clinical setting regardless of risk — has been examined as one response to this burden, but evidence for its effectiveness remains limited (36, 37). The U.S. Preventative Services Task Force (USPSTF) has concluded that evidence is insufficient to recommend for or against universal suicide risk screening in the adult population, citing in part inadequate evidence on the accuracy of available screening tools (36, 37). Standard suicide screening tools have historically not been validated for use in ASD populations, where communication differences and atypical symptom presentation complicate standard assessment (38, 39). Moreover, clinicians report significantly lower self-efficacy in assessing suicide risk in individuals with ASD compared to clients without ASD (39). These limitations suggest that universal screening approaches alone are insufficient to address the elevated SA burden in ASD (11, 39, 40).

What is needed instead is a risk stratification approach — defined as the process of classifying patients according to their health risks and projected care needs, enabling clinical resources and monitoring to be directed toward those at highest risk rather than applied uniformly across all patients (41). Unlike universal screening, which aims to identify any individual who may be at risk, risk stratification leverages population-level data on demographic and clinical characteristics to identify which subgroups carry the greatest burden, supporting a more targeted and resource-efficient approach to prevention (41, 42). In the context of suicide prevention, EHR-based risk stratification approaches have demonstrated particular promise in general psychiatric and veteran populations — machine learning models applied to structured EHR data have markedly outperformed clinician-led risk assessments, and programs such as the VA Recovery Engagement and Coordination for Health-Veterans Enhanced Treatment (REACH-VET) initiative have demonstrated that targeting enhanced outreach to the highest-risk tier of patients can meaningfully reduce documented suicide attempts (42–44). However, existing EHR-based suicide risk models have been developed primarily in general psychiatric and veteran populations, and the clinical features that predict SA risk in ASD differ meaningfully from those in the general population. ASD-specific factors including trauma symptoms, sensory sensitivities, and distinctive comorbidity patterns are important predictors of SA risk in individuals with ASD that would not be captured by general population models (11, 45). Developing effective risk stratification approaches for ASD therefore requires population-level prevalence data characterizing how SA risk is distributed across demographic and clinical subgroups specific to this population — data that have been largely unavailable at the scale and granularity needed (6, 9). While previous studies have explored the relationship between psychiatric diagnosis and SA in ASD, much of this work has been limited by small sample sizes or single-country datasets (8, 9, 12, 20), and the role of specific substance use disorder subtypes in conferring differential SA risk has not been characterized at population scale.

The present study addresses this need using a large national electronic health record dataset comprising over 2.3 million individuals with ASD — far exceeding the sample sizes of prior population-based studies examining SA in this population, including a Danish population-based cohort of 35,020 individuals with ASD (20), a Swedish national register study of 54,168 individuals with ASD (8), and a Canadian population-based matched cohort of 75,926 individuals with ASD (13). We characterize SA prevalence when stratified by sex, age group, and psychiatric comorbidity type, with particular attention to substance use disorder subtypes — demographic and clinical dimensions that have not been simultaneously characterized in ASD and that are directly relevant to identifying high-risk subgroups for targeted clinical intervention (11, 45). Through this characterization, complemented by supplementary sensitivity analyses examining SA prevalence by intellectual disability status and race/ethnicity, we provide a population-level descriptive foundation needed to inform evidence-based suicide risk stratification in ASD — establishing which subgroups carry the greatest SA burden and supporting more targeted, efficient, and clinically meaningful approaches to suicide prevention in this high-risk population.

2 Materials and methods

2.1 Study design and data source

We conducted a retrospective, cross-sectional study using deidentified national EHR data from the Epic Cosmos data warehouse (46). Epic Cosmos aggregates EHRs from more than 2,133 hospitals, 48,800 clinics, and 304 million unique patients, representing geographically and demographically diverse populations across inpatient, outpatient, and emergency settings. Participating health systems agree to contribute their data, which is pooled and de-identified at source. Each patient has a unique identifier, and their inpatient and outpatient charts are integrated into a single comprehensive record that moves with them among constituent healthcare systems. Cosmos data are curated through a multi-stage process managed jointly by Epic and participating health systems. Each site securely transmits its data after mapping local values to standardized vocabularies before transfer. Epic conducts a comprehensive validation to verify data parity with the source systems, corrects mapping errors, and assesses completeness before approving data for inclusion. All workflows were conducted within the Cosmos Data Science Virtual Machine (DSVM) environment, which prohibits the export of any line-level or patient-level data to ensure compliance with data privacy and security standards. The analytic period spanned January 1, 2016, through December 31, 2025.

2.2 Study population

The study population included patients aged five years or older with a lifetime diagnosis of autism spectrum disorder (ASD). ASD diagnoses were identified using the following ICD-10-CM codes: F84.0 (Autistic disorder), F84.5 (Asperger's syndrome), F84.8 (Other pervasive developmental disorders), and F84.9 (Pervasive developmental disorder, unspecified). Individuals were stratified by sex (female, male) and age group (5–14, 15–24, 25–34, 35–44, 45–64, and 65+ years). Persons whose sex was neither male nor female were excluded.

Race and ethnicity were combined into a single variable using the most frequently recorded non-missing value across all visits (mode), excluding missing/unknown values from the mode calculation; the most recent non-missing value was used as a tiebreaker in cases of ties, and patients with no non-missing value recorded were classified as missing/unknown. Race/ethnicity was stratified into nine categories: Hispanic or Latino (any race), Non-Hispanic White, Non-Hispanic Black or African American, Non-Hispanic Asian, Non-Hispanic Multi-race, Non-Hispanic Other Race, Non-Hispanic American Indian/Alaska Native or Native Hawaiian/Pacific Islander (combined), Non-Hispanic Missing/Unknown Race, and Missing/Unknown. American Indian/Alaska Native and Native Hawaiian/Pacific Islander individuals were combined into a single category given their small individual population sizes within this cohort. Small-population groups are frequently suppressed or aggregated in health research due to disclosure limitation requirements, a practice that can further marginalize already underrepresented populations by excluding them from reported findings entirely; combining these groups into a labeled category, rather than suppressing them individually, was intended to preserve their representation in this analysis (47).

Intellectual disability (ID) status was defined as a binary indicator based on the presence of any ICD-10-CM code for intellectual disability (F70-F73, F78, F78.A, F78.A1, F78.A9, F79) at any point in the patient's record; patients without any such code were classified as not having ID.

Psychiatric comorbidities were classified using ICD-10-CM codes and included eating disorders, mood disorders, schizophrenia and psychotic disorders, substance-related disorders, and trauma and stressor-related disorders (Supplementary Table 1). Substance-related disorders were further stratified by subtype: alcohol use disorder, opioid use disorder, cannabis use disorder, stimulant use disorder, and other substances (Supplementary Table 1). Individuals with multiple psychiatric diagnoses were included in each relevant comorbidity category.

Person-first language ("individuals with ASD") is used throughout this manuscript, consistent with American Psychological Association style guidance emphasizing the person over the diagnosis (48). We recognize that language preferences within the ASD community are not uniform, with some self-advocates preferring identity-first terminology (49); our use of person-first language reflects a deliberate stylistic choice rather than an assertion that this convention is universally preferred.

2.3 Outcome definition

The primary outcome was a recorded suicide attempt during the study period, identified using ICD-10-CM codes for intentional self-harm (Supplementary Table 2). This code set corresponds to the Centers for Disease Control and Prevention's surveillance case

definition for nonfatal suicide attempts (50), the standard definition used in administrative-data suicide research. R45.88 (self-harm without suicidal intent), the ICD-10-CM code specifically designated to distinguish non-suicidal self-injury from suicidal self-harm, was not included in this code set. The outcome reported here therefore reflects recorded EHR-coded suicide attempt/self-harm events rather than confirmed suicide attempts verified by clinical or forensic review. Administrative coding of suicide attempts using ICD-10-CM codes is known to undercount true prevalence, as sensitivity for identifying nonfatal suicidal behaviors in administrative data is low and varies substantially across settings and populations (51). SA prevalence was calculated as the number of unique patients with at least one recorded suicide attempt divided by the total number of patients in each stratum, expressed as a percentage with 95% confidence intervals calculated using the Wilson score method (52).

2.4 Statistical analysis

Descriptive statistics were used to characterize the study population and calculate crude prevalence of suicide attempt across all strata. Modified Poisson regression with robust variance estimation was used to calculate adjusted prevalence ratios (aPRs) and 95% confidence intervals, with history of suicide attempt as the binary outcome. This approach was selected over logistic regression as prevalence ratios are the more appropriate measure of association in cross-sectional studies with binary outcomes, providing more direct and interpretable estimates than odds ratios (53, 54). Three models were fit (1): an overall model including sex, age group, and presence of at least one psychiatric comorbidity as covariates (2); sex-stratified models to assess whether associations differed between females and males; and (3) separate unadjusted models for each psychiatric comorbidity type and substance use disorder subtype compared to no comorbidity. Comorbidity-specific prevalence ratios are presented unadjusted as the aggregated nature of the data precluded simultaneous adjustment for multiple psychiatric comorbidities, given that individuals may carry more than one diagnosis. Reference categories were male sex, age group 5–14, and no psychiatric comorbidity. Statistical significance was defined as $p < 0.05$. All analyses were conducted using Python (version 3.13) with the statsmodels library (55). Cell counts of 10 or fewer were suppressed in accordance with Epic Cosmos data privacy policies and excluded from analyses. For sensitivity analyses involving intellectual disability and race/ethnicity, prevalence estimates affected by cell suppression were reported as a bounded range rather than a single value: the lower bound reflects only non-suppressed cells, and the upper bound assumes each suppressed cell contained the maximum possible suppressed value of ten (56).

2.5 Ethics

This study was determined to not meet the definitions of Human Subjects Research by the University of Utah Institutional Review Board (IRB_00202177) and was therefore exempt from informed consent requirements.

2.6 Data availability statement

The data used in this study are derived from the Epic Cosmos data warehouse and are not publicly available due to data privacy and security restrictions inherent to the Cosmos Data Science Virtual Machine (DSVM) environment. Researchers interested in accessing Epic Cosmos data should contact Epic Systems or their participating institution. The analytic code used in this study is available from the de Lacy lab's GitHub (delacylab/ASD_Suicide_Prevalence_EHR_Data).

3 Results

3.1 Sample characteristics

A total of 2,311,171 individuals with ASD were identified in the Epic Cosmos database during the study period (2016–2025), of whom 629,895 (27.3%) were female and 1,681,276 (72.7%) were male. Overall, 38,160 individuals (1.7%) had at least one recorded suicide attempt during the study period. The age distribution of the sample was as follows: 5–14 years (44.1%), 15–24 years (29.5%), 25–34 years (15.8%), 35–44 years (5.6%), 45–64 years (3.6%), and 65+ years (1.4%). A total of 646,643 individuals (28.0%) had at least one recorded psychiatric comorbidity, with females more likely to have a comorbidity (36.8%) than males (24.7%). Full sample characteristics are presented in Table 1. SA prevalence by sex and age group is examined in the sections that follow.

3.2 Suicide attempt prevalence by sex

Females with ASD had considerably higher SA prevalence than males (2.9% vs 1.2%; 18,158 vs 20,002 recorded suicide attempts), and this difference remained significant after adjustment for age group and psychiatric comorbidity (aPR 1.61, 95% CI 1.35–1.92, p = 1.07e-07), representing a 61% higher prevalence in females; overall adjusted prevalence ratios are presented in Figure 1. Sex-stratified analyses examining age and comorbidity effects separately are presented in Table 2 and described in the sections below.

3.3 Suicide attempt prevalence by age

SA prevalence varied markedly across age groups, peaking in the 15–24 age group and declining progressively with advancing age. Crude SA prevalence was 0.23% in the 5–14 reference group, rising to 2.79% in the 15–24 age group, 3.12% in the 25–34 age group, 2.60% in the 35–44 age group, 2.09% in the 45–64 age group, and 0.99% in the 65+ age group. After adjustment, all age groups showed significantly elevated SA prevalence relative to the 5–14 reference group except the 65+ age group: 15–24 (aPR 5.14, 95% CI 3.62–7.29, p = 5.57e-20), 25–34 (aPR 4.46, 95% CI 3.10–6.42, p = 7.45e-16), 35–44 (aPR 3.09, 95% CI 2.01–4.76, p = 2.76e-07), 45–64 (aPR 2.39, 95% CI 1.63–3.50, p = 7.46e-06), and 65+ (aPR 1.23, 95% CI 0.83–1.80, p = 0.299). Full adjusted prevalence ratios by age group are presented in Table 3.

TABLE 1 Sample characteristics of individuals with ASD, stratified by sex.

Characteristic	Female N (%)	Male N (%)	Total N (%)
Total patients	629,895 (27.3%)	1,681,276 (72.7%)	2,311,171 (100%)
Suicide attempt	18,158 (2.9%)	20,002 (1.2%)	38,160 (1.7%)
Age group			
5–14	257,403 (40.9%)	762,828 (45.4%)	1,020,231 (44.1%)
15–24	180,519 (28.7%)	500,320 (29.8%)	680,839 (29.5%)
25–34	106,511 (16.9%)	259,078 (15.4%)	365,589 (15.8%)
35–44	45,171 (7.2%)	84,883 (5.0%)	130,054 (5.6%)
45–64	29,531 (4.7%)	52,650 (3.1%)	82,181 (3.6%)
65+	10,760 (1.7%)	21,517 (1.3%)	32,277 (1.4%)
Comorbidity			
At least one	232,074 (36.8%)	414,569 (24.7%)	646,643 (28.0%)
None	397,821 (63.2%)	1,266,707 (75.3%)	1,664,528 (72.0%)

Counts and percentages are reported for females, males, and the total sample.
Demographic and clinical characteristics of 2,311,171 individuals with ASD identified in the Epic Cosmos database (2016–2025).

3.4 Suicide attempt prevalence by age and sex

Age patterns differed meaningfully by sex. Among females, crude SA prevalence peaked in the 15–24 age group (5.40%) and declined progressively across older groups (25–34: 4.60%; 35–44: 3.19%; 45–64: 2.62%; 65+: 1.23%), with an adjusted peak at 15–24 years (aPR 4.33, 95% CI 4.23–4.43). Among males, crude SA prevalence was lower across all age groups, peaking at 25–34 years (2.51%) rather than 15–24 years (1.84%), with an adjusted peak similarly at 25–34 years (aPR 6.08, 95% CI 5.05–7.31). Notably, females aged 65+ showed a significantly lower SA prevalence compared to the 5–14 reference group (aPR 0.80, 95% CI 0.78–0.82), while males aged 65+ remained at elevated risk (aPR 1.71, 95% CI 1.44–2.04, p = 1.52e-09). Sex-stratified age results are presented in Table 2 and Figure 2.

3.5 Suicide attempt prevalence by psychiatric comorbidity

Having at least one psychiatric comorbidity was associated with markedly higher SA prevalence compared to having no comorbidity. Among individuals with at least one psychiatric comorbidity, crude SA prevalence was 7.49% compared to 0.21% in those with no comorbidity, representing a 30-fold difference after adjustment (aPR 30.56, 95% CI 26.03–35.89, p < 2.20e-16). Full adjusted prevalence ratios are presented in Table 3.

SA prevalence varied substantially across comorbidity types (Table 4). Substance-related disorders were associated with the highest crude SA prevalence (16.9%) and the highest unadjusted prevalence ratio among comorbidity categories (uPR 134.20, 95% CI 67.68–266.10, p < 2.20e-16), followed by schizophrenia and psychotic disorders (crude prevalence 11.8%; uPR 93.89, 95% CI

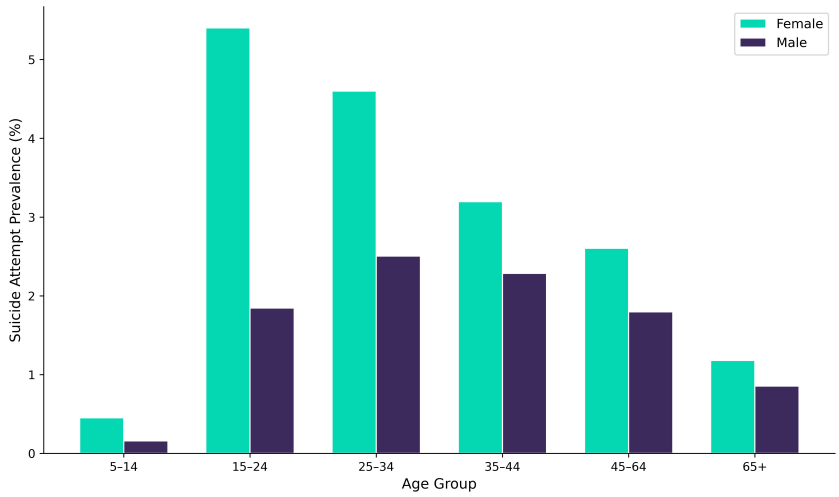


FIGURE 1
Suicide attempt prevalence by sex and age group in individuals with autism spectrum disorder. Bars represent crude prevalence (%) of recorded suicide attempt within each age group, stratified by sex.

10.3389/fpsyt.2026.1940193

TABLE 2 Adjusted prevalence ratios (aPRs) for suicide attempt in individuals with ASD: sex-stratified models.

Variable	Female aPR	Female 95% CI	Female p-value	Male aPR	Male 95% CI	Male p-value
Age group						
5–14 (ref)	—	—	—	—	—	—
15–24	4.33	4.23–4.43	< 2.20e-16	5.74	4.81–6.85	< 2.20e-16
25–34	3.02	2.92–3.12	< 2.20e-16	6.08	5.05–7.31	< 2.20e-16
35–44	1.92	1.88–1.95	< 2.20e-16	4.60	3.84–5.51	< 2.20e-16
45–64	1.58	1.55–1.62	< 2.20e-16	3.35	2.81–3.99	< 2.20e-16
65+	0.80	0.78–0.82	< 2.20e-16	1.71	1.44–2.04	1.52e-09
Comorbidity						
At least one	36.54	31.04–43.01	< 2.20e-16	27.77	22.65–34.06	< 2.20e-16
None (ref)	—	—	—	—	—	—

ASD, autism spectrum disorder; Apr, adjusted prevalence ratio; CI, confidence interval; ref, reference category. Modified Poisson regression with robust variance estimation (HC3). Cell counts of ≤10 suppressed per Epic Cosmos data privacy policies and excluded from analysis.
Sex-stratified adjusted prevalence ratios from modified Poisson regression with robust variance estimation, with age group 5–14 years and no psychiatric comorbidity as reference categories.

47.21–186.72, p < 2.20e-16). Trauma and stressor-related disorders (8.2%), mood disorders (7.4%), and eating disorders (5.0%) showed lower but still considerably elevated crude prevalences; unadjusted prevalence ratios for all comorbidity types are presented in Table 5. Substance-related disorders were further stratified by subtype; subtype-specific prevalence estimates and unadjusted prevalence ratios are presented in the following section.

The effect of psychiatric comorbidity differed by sex. Among females with at least one psychiatric comorbidity, crude SA prevalence was 9.3% compared to 0.45% in females with no comorbidity (aPR 36.54, 95% CI 31.04–43.01, p < 2.20e-16). Among males, crude SA prevalence was 6.2% with comorbidity compared to 0.16% without (aPR 27.77, 95% CI 22.65–34.06, p < 2.20e-16). Sex-stratified comorbidity results are presented in Table 2.

TABLE 3 Adjusted prevalence ratios (aPRs) for suicide attempt in individuals with ASD: overall model.

Variable	aPR	95% CI	p-value
Sex			
Female	1.61	1.35–1.92	1.07e-07
Male (ref)	—	—	—
Age group			
5–14 (ref)	—	—	—
15–24	5.14	3.62–7.29	5.57e-20
25–34	4.46	3.10–6.42	7.45e-16
35–44	3.09	2.01–4.76	2.76e-07
45–64	2.39	1.63–3.50	7.46e-06
65+	1.23	0.83–1.80	2.99e-01
Comorbidity			
At least one	30.56	26.03–35.89	< 2.20e-16
None (ref)	—	—	—

Adjusted prevalence ratios from modified Poisson regression with robust variance estimation, including sex, age group, and presence of at least one psychiatric comorbidity as covariates. Reference categories: male sex, age group 5–14 years, no psychiatric comorbidity.

Crude SA prevalence among individuals with at least one psychiatric comorbidity was elevated across all age groups, peaking in the 15–24 age group (7.86% with comorbidity vs 0.26% without) and declining progressively with advancing age. Age-stratified comorbidity prevalence data are presented in Supplementary Table 3.

3.6 Suicide attempt prevalence by substance use disorder subtype

Among individuals with substance-related disorders, SA prevalence varied across subtypes. Stimulant use disorder was associated with the highest crude SA prevalence (24.3%) and the highest unadjusted prevalence ratio of any subtype examined (uPR 196.11, 95% CI 100.63–382.20, p < 2.20e-16), followed by other substances (crude prevalence 21.3%; uPR 177.89, 95% CI 91.29–346.64, p < 2.20e-16), opioid use disorder (19.7%; uPR 174.59, 95% CI 88.56–344.19, p < 2.20e-16), cannabis use disorder (21.4%; uPR 170.70, 95% CI 87.30–333.78, p < 2.20e-16), and alcohol use disorder (19.9%; uPR 158.71, 95% CI 80.18–314.16, p < 2.20e-16). All substance use disorder subtypes showed substantially higher unadjusted prevalence ratios than the broader substance-related disorders category (uPR 134.20). Full subtype-specific results are presented in Table 5.

Crude SA prevalence across all substance use disorder subtypes was highest in the 15–24 age group and declined progressively with advancing age. Stimulant use disorder showed the highest crude SA prevalence in the 15–24 age group (35.6%), followed by opioid use disorder (31.0%), alcohol use disorder (30.1%), other substances (29.1%), and cannabis use disorder (26.7%). Age-stratified subtype prevalence data are presented in Supplementary Table 3.

SA prevalence across subtypes was consistently higher in females than males. Among females, stimulant use disorder showed the highest crude prevalence (27.1%; uPR 178.26, 95% CI 71.39–445.16, p < 2.20e-16), followed by cannabis use disorder (26.1%; uPR 167.56, 95% CI 67.73–414.54, p < 2.20e-16) and alcohol use disorder (25.0%; uPR 161.36, 95% CI 64.45–403.97, p < 2.20e-16). Among males, stimulant use disorder again showed the highest unadjusted prevalence ratio (uPR 198.75, 95% CI 86.60–

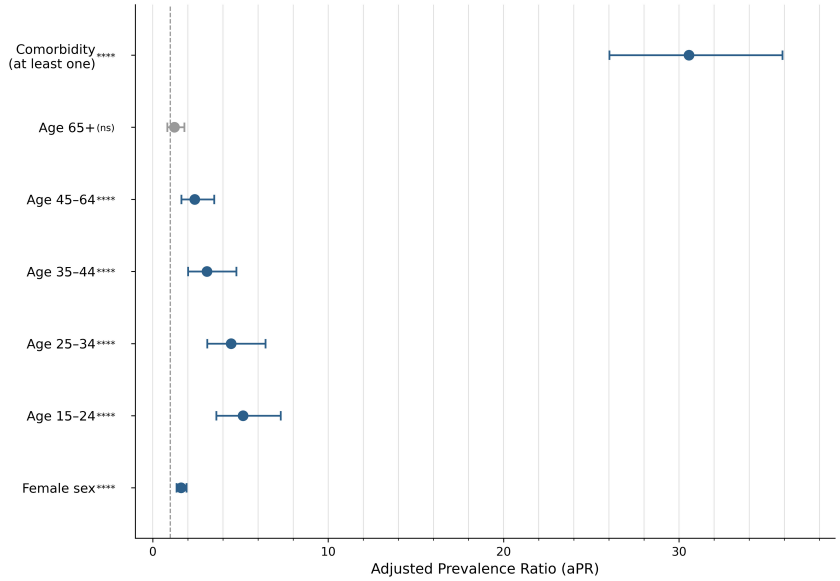


FIGURE 2
Adjusted prevalence ratios (aPRs) for suicide attempt in individuals with autism spectrum disorder: overall model. Points represent aPRs with 95% confidence intervals from modified Poisson regression with robust variance estimation. Reference categories: male sex, age group 5–14 years, no psychiatric comorbidity. Asterisks denote statistical significance: * p<0.05, ** p<0.01, *** p<0.001, **** p<0.0001; ns, not significant. Gray point indicates the non-significant association.

456.14, p < 2.20e-16), followed by opioid use disorder (uPR 192.49, 95% CI 82.43–449.47, p < 2.20e-16) and other substances (uPR 183.75, 95% CI 80.23–420.85, p < 2.20e-16). Sex-stratified subtype results are presented in Supplementary Table 4.

3.7 Suicide attempt prevalence by intellectual disability status and race/ethnicity (sensitivity analyses)

Supplementary sensitivity analyses examined SA prevalence by intellectual disability (ID) status and by race/ethnicity, using a data extraction obtained subsequent to the primary analysis (N = 2,576,809 vs. N = 2,311,171). Prevalence estimates affected by cell suppression are reported as bounded ranges. Overall SA prevalence in this updated cohort was 1.66–1.68%. SA prevalence was 1.58% among individuals without ID and 2.37–2.38% among individuals with ID. SA prevalence by race/ethnicity was as follows: Hispanic or Latino (any race), 0.78–0.80%; Missing/Unknown, 0.91–0.93%; Non-Hispanic American Indian/Alaska Native or Native Hawaiian/Pacific Islander (combined), 1.07–1.98%; Non-Hispanic Asian, 0.70–0.84%; Non-Hispanic Black or African American, 1.23–1.26%; Non-Hispanic Missing/Unknown Race, 0.72–1.21%; Non-Hispanic Multi-race, 2.25–2.28%; Non-Hispanic Other Race, 0.65–0.92%; and Non-Hispanic White, 2.22%.

Comorbidity-type unadjusted prevalence ratios (uPR) stratified by ID and by race/ethnicity are presented in Supplementary Table 5

TABLE 4 Prevalence of suicide attempt by psychiatric comorbidity, stratified by sex.

Comorbidity	Female N	Female SA, N (%)	Male N	Male SA, N (%)	Total N	Total SA, N (%)
Eating Disorders	30,707	2,934 (9.6%)	47,142	931 (2%)	77,849	3,865 (5.0%)
Mood Disorders	184,917	17,130 (9.3%)	284,808	17,667 (6.2%)	469,725	34,797 (7.4%)
Schizophrenia and Psychotic Disorders	28,794	4,591 (15.9%)	61,566	6,071 (9.9%)	90,360	10,662 (11.8%)
Substance-Related Disorders	21,021	4,420 (21%)	42,007	6,210 (14.8%)	63,028	10,630 (16.9%)
Alcohol Use Disorder	7,732	1,935 (25%)	18,324	3,239 (17.7%)	26,056	5,174 (19.9%)
Opioid Use Disorder	3,627	705 (19.4%)	5,954	1,187 (19.9%)	9,581	1,892 (19.7%)
Cannabis Use Disorder	9,242	2,413 (26.1%)	18,887	3,603 (19.1%)	28,129	6,016 (21.4%)
Stimulant Use Disorder	4,258	1,153 (27.1%)	8,788	2,018 (23%)	13,046	3,171 (24.3%)
Other Substances	7,302	1,768 (24.2%)	15,359	3,061 (19.9%)	22,661	4,829 (21.3%)
Trauma and Stressor-Related Disorders	91,206	10,259 (11.2%)	139,123	8,620 (6.2%)	230,329	18,879 (8.2%)

SA, suicide attempt. Patients may appear in multiple comorbidity rows as diagnoses are not mutually exclusive. Substance-related disorder subtypes are indented below the parent category. Cell counts of ≤10 suppressed per Epic Cosmos data privacy policies and excluded from analyses.
Sample sizes and crude suicide attempt prevalence for each psychiatric comorbidity category and substance use disorder subtype, reported for females, males, and the total sample. Substance use disorder subtypes are indented below the parent category. Individuals with multiple diagnoses are included in each relevant category.

TABLE 5 Unadjusted prevalence ratios (uPRs) for suicide attempt by psychiatric comorbidity type and substance use disorder subtype in individuals with ASD.

Comorbidity	uPR	95% CI	p-value
Psychiatric comorbidity type			
Eating Disorders	39.78	12.12–130.60	1.25e-09
Mood Disorders	58.95	29.35–118.40	< 2.20e-16
Schizophrenia and Psychotic Disorders	93.89	47.21–186.72	< 2.20e-16
Substance-Related Disorders	134.20	67.68–266.10	< 2.20e-16
Trauma and Stressor-Related Disorders	65.22	29.32–145.09	< 2.20e-16
Substance use disorder subtype			
Alcohol Use Disorder	158.71	80.18–314.16	< 2.20e-16
Opioid Use Disorder	174.59	88.56–344.19	< 2.20e-16
Cannabis Use Disorder	170.70	87.30–333.78	< 2.20e-16
Stimulant Use Disorder	196.11	100.63–382.20	< 2.20e-16
Other Substances	177.89	91.29–346.64	< 2.20e-16

Each comorbidity type and substance use disorder subtype was compared separately against individuals with no psychiatric comorbidity as the reference group. uPRs are presented unadjusted, as the aggregated nature of the data precluded simultaneous adjustment for multiple comorbidities.

and 6, respectively. Consistent with the primary analysis, stimulant use disorder showed the highest unadjusted prevalence ratio in both ID groups (uPR 193.94 without ID; uPR 197.58 with ID) and in the majority of race/ethnicity categories with reportable comorbidity-type estimates. Several race/ethnicity categories (Non-Hispanic AI/AN or NH/PI [combined], Non-Hispanic Missing/Unknown Race, Non-Hispanic Asian, and Non-Hispanic Other Race) had insufficient cell counts to support reliable uPR estimates for some comorbidity types; excluded estimates are noted directly in the supplementary tables.

4 Discussion

This study characterized suicide attempt prevalence in the largest real-world cohort of individuals with ASD to date, comprising 2,311,171 individuals drawn from the Epic Cosmos database — far exceeding the sample sizes of prior population-based studies, including a Swedish national register study of 54,168 individuals with ASD (8) and a Canadian population-based matched cohort of 75,926 individuals with ASD (13), the latter of which examined self-harm events broadly rather than suicide attempts specifically. This sample size enables stable prevalence estimates across multiple demographic and clinical subgroups simultaneously, stratified by sex, age group, and psychiatric comorbidity — providing the population-level descriptive foundation needed to inform evidence-based suicide risk stratification in ASD. Our findings demonstrate that SA prevalence is elevated relative to the general population in ASD, particularly among those with psychiatric comorbidities, and that sex and age patterns of SA prevalence differ meaningfully between females and males.

The overall SA prevalence of 1.7% in our sample is lower than estimates from previous studies. A recent systematic review reported lifetime SA prevalence of 15.3% and 12-month SA prevalence of 14.1% in ASD (9), while past-year SA prevalence in the general US adult population is approximately 0.8% (57). The lower prevalence observed in the current study likely reflects key methodological differences — our cross-sectional analysis captures SA recorded within a fixed EHR observation window using ICD-10-CM diagnostic codes, rather than self-reported lifetime or 12-month history, which tends to yield higher estimates (6, 9). Additionally, administrative coding of SA is known to undercount true prevalence, as ICD-10-CM codes demonstrate limited sensitivity for identifying nonfatal suicidal behaviors and only attempts resulting in a documented clinical encounter within participating health systems are captured (51). Despite these methodological differences, our observed prevalence of 1.7% overall exceeds general population estimates and is consistent with the elevated SA burden documented in prior ASD research (8, 9). However, aggregate prevalence figures alone obscure the variation in SA burden across demographic and clinical subgroups. Prevalence rising to 16.9% in those with substance-related disorders and 11.8% in those with schizophrenia and psychotic disorders illustrates the concentration of burden within specific subgroups — and why population-level subgroup characterization is essential for developing targeted risk stratification approaches in ASD (9, 58).

4.1 Sex differences in suicide attempt prevalence

Females with ASD have significantly higher SA prevalence than males (2.9% vs. 1.2%; aPR 1.61, 95% CI 1.35–1.92), a finding consistent with an evolving literature in which sex differences in ASD suicidality have only recently come into focus (6, 9, 18, 19). Earlier population-based studies found no significant sex difference in SA prevalence in ASD (34, 59), while more recent large population-based studies — including a Danish population-based cohort of 35,020 individuals with ASD (20) and a Swedish national register study of 54,168 individuals with ASD (8) — identified higher SA prevalence in females with ASD than males, a pattern subsequently confirmed in systematic reviews and meta-analyses (6, 9, 19). The emergence of this sex difference in more recent and larger studies likely reflects improved diagnostic recognition of ASD in females over the past two decades — ASD has historically been characterized as a predominantly male condition, with foundational research skewed toward male presentations and females consistently underdiagnosed or diagnosed later (18, 21, 60, 61), and prior estimates drawn from cohorts with lower female representation are likely to have underestimated sex differences in SA prevalence (18, 19). The sex difference in SA prevalence in ASD mirrors the pattern observed in the general population, where females also report higher rates of SA than males (57, 62), though the mechanisms associated with elevated SA prevalence in females with ASD are likely distinct from those in the general population, reflecting ASD-specific vulnerabilities — including camouflaging, delayed diagnosis, and elevated psychiatric comorbidity burden — documented across qualitative and quantitative research over the past decade (10, 11, 19, 45, 63). It is important to note that this sex difference in SA prevalence does not extend to suicide death — studies of suicide

mortality in ASD have consistently found comparable rates between males and females with ASD (8, 18) — a divergence that underscores the importance of examining SA and suicide death as conceptually and clinically distinct outcomes, and one that is obscured when suicidal behaviors are aggregated, as has been common in prior research (6, 9, 14, 15, 18).

Several converging ASD-specific vulnerabilities may account for the elevated SA prevalence observed in females with ASD relative to males. Camouflaging — the conscious or unconscious suppression of autistic traits to conform to social expectations — is more prevalent in females with ASD than males and is directly associated with suicidality, likely through its links to psychological defeat and internal entrapment (10, 19). Females with ASD face greater gendered social expectations to socialize and conform, generating higher camouflaging demands and the sustained psychological costs of inauthenticity across social contexts (10, 19, 63). Diagnostic delay compounds this burden — females with ASD are consistently diagnosed later than males on average (21, 61, 64), which may contribute to prolonged periods of unrecognized struggle, unmet support needs, and potential misattribution of distress before ASD identification occurs (22, 63, 64). Elevated psychiatric comorbidity burden in females with ASD further compounds vulnerability: in our sample, 36.8% of females had at least one psychiatric comorbidity compared to 24.7% of males, and the association between comorbidity and SA prevalence was stronger in females (aPR 36.54) than in males (aPR 27.77), consistent with evidence that females with ASD have higher rates of internalizing disorders than males with ASD (9, 12, 20). These converging vulnerabilities — greater camouflaging demands, later diagnosis, and higher psychiatric comorbidity burden — likely interact and may collectively contribute to the elevated SA prevalence observed in females with ASD relative to males, though the relative contributions of each remain incompletely characterized (11, 18).

The age pattern of SA prevalence differs meaningfully between sexes. Among females with ASD, SA prevalence peaks in the 15–24 age group (aPR 4.33, 95% CI 4.23–4.43), while among males it peaks later at 25–34 years (aPR 6.08, 95% CI 5.05–7.31). This sex-specific divergence in the age of peak SA prevalence has not previously been documented at population scale in ASD, as prior studies have generally lacked the sample sizes needed to examine age-by-sex interactions in SA specifically (6, 9, 18). The earlier peak in females is consistent with the intensification of social demands during adolescence and early adulthood — the period when camouflaging becomes most cognitively and emotionally costly, when undiagnosed females with ASD are most likely to be struggling without adequate support, and when the gap between social expectations and autistic experience may be most acutely felt (19, 22, 23, 63, 64). The later peak in males at 25–34 years may reflect the accumulation of stressors associated with post-educational transitions — including high rates of unemployment, difficulties with independent living, and social isolation — as males with ASD enter adulthood; social isolation in particular has been linked to thwarted belongingness, a core construct of the interpersonal theory of suicide (22, 29, 65). Within the interpersonal theory of suicide, these experiences map onto thwarted belongingness and perceived burdensomeness, both of which have been specifically implicated in ASD suicidality and may accumulate gradually across the transition to adulthood in males rather than manifesting acutely in adolescence (29, 65). These sex-specific age patterns suggest that clinical attention in suicide prevention efforts should be directed toward females with ASD in the 15–24 age window and males with ASD across a broader range extending into the mid-twenties and beyond.

The lower SA prevalence in females with ASD aged 65+ relative to the 5–14 reference group (aPR 0.80, 95% CI 0.78–0.82) — in contrast to males aged 65+ who remain at elevated risk (aPR 1.71, 95% CI 1.44–2.04) — is an interesting finding that likely reflects multiple non-mutually-exclusive processes. Selective survival is one plausible contributor: females with ASD who reach older age may represent a subgroup with lower baseline vulnerability, a phenomenon identified as an important consideration in studies of older adults with neurodevelopmental disorders more broadly (66). Cohort effects are likely also relevant — older females with ASD in our sample were far less likely to have received formal ASD recognition during their lifetime given historically low diagnostic rates in females (21, 60, 61), and this systematic under identification would suppress documented SA rates through reduced clinical contact and diagnostic coding (25, 26, 51). The persistently elevated SA prevalence in older males with ASD (aPR 1.71) is consistent with the well-documented vulnerability of older males to suicide-related outcomes in the general population, where social isolation following relationship loss or bereavement, declining physical health, and reduced help-seeking behavior converge to sustain risk into late life (57, 62). In ASD, these general population patterns may be compounded by the cumulative burden of lifelong social exclusion and the relative absence of established social networks that might otherwise serve as protective factors in older adulthood — consistent with emerging evidence that suicidality in adults with ASD does not follow the same age-related decline observed in the general population and may even increase with age in some groups (24–26).

4.2 Age differences in suicide attempt prevalence

SA prevalence peaks in the 15–24 age group (aPR 5.14, 95% CI 3.62–7.29) and declines progressively with advancing age, though it remains significantly elevated relative to the 5–14 reference group across all age groups except 65 +. This overall age pattern is broadly consistent with prior population-based data showing that adolescence and young adulthood represent a period of heightened SA vulnerability in ASD (20, 22, 23). The mechanisms underlying this peak are likely multifactorial, reflecting the confluence of increasing social demands, diagnostic transitions, loss of structured educational support, and the onset or exacerbation of psychiatric comorbidities that converge during this developmental window (11, 22, 23). ADHD — among the most common comorbidities in ASD (5) — may be a particularly important contributor to elevated SA risk during adolescence and young adulthood, given that ADHD-associated impulsivity, emotional dysregulation, and executive function deficits have been independently associated with SA risk in adolescent populations, and untreated ADHD in individuals with ASD has been specifically linked to elevated suicidality in late adolescence and early adulthood (5, 22, 23). Formal support for individuals with ASD often declines during this period —

service reductions frequently begin before high school graduation and continue thereafter, creating a window of heightened vulnerability with diminished institutional scaffolding precisely when the demands of adult life are increasing (29, 65, 67).

The progressive decline in SA prevalence from the 25–34 age group onward — with aPRs of 4.46 (25–34), 3.09 (35–44), and 2.39 (45–64) — is broadly consistent with the general population trend of declining SA prevalence with advancing age (57, 62). However, SA prevalence in our ASD sample exceeds general population estimates across all age groups, and individuals with ASD who die by suicide have a mean age at death of 32.4 years — considerably younger than the non-ASD average of 41.8 years — suggesting that suicide risk in ASD extends well into midlife rather than concentrating exclusively in adolescence (68). The persistently elevated SA prevalence in the 35–44 and 45–64 age groups — despite the progressive decline — likely reflects an accumulation of chronic stressors common among adults with ASD during midlife, including sustained high rates of unemployment, social isolation, relationship difficulties, and the cumulative burden of untreated or undertreated psychiatric comorbidities (11). Within the interpersonal theory of suicide, social isolation is understood as a driver of thwarted belongingness, one of the framework's core constructs (29, 65). These findings add population-scale evidence to an important but undercharacterized dimension of ASD suicide research — the persistence of elevated risk across the adult lifespan — a gap that has been highlighted given that research on middle-aged and older adults with ASD accounts for only 0.4% of indexed autism research (24–26).

The non-significant association in the 65+ age group overall (aPR 1.23, 95% CI 0.83–1.80, p = 0.299) warrants careful interpretation rather than straightforward conclusion that SA risk attenuates in older adults with ASD. Several non-mutually-exclusive explanations are plausible. Systematic under identification of ASD in older cohorts is likely a major contributor — estimates suggest that approximately nine in ten adults with ASD aged 50 and over in the UK remain undiagnosed or misdiagnosed (26), and this suppression of ASD ascertainment would similarly reduce the likelihood of SA documentation in clinical records (25, 51). Survivor bias may also contribute, whereby older individuals with ASD who remain engaged with the healthcare system represent a subgroup with lower baseline vulnerability than those who have died prematurely or disengaged from care (66). The sex-specific breakdown of this finding — with females aged 65+ showing significantly lower SA prevalence than the reference group (aPR 0.80) while males aged 65+ remain at elevated risk (aPR 1.71) — is discussed in the context of cohort effects, diagnostic recognition, and selective survival in Section 4.1 above, and illustrates why the overall 65+ estimate should not be interpreted without this stratification.

Emerging evidence raises important questions about whether SA risk truly attenuates with age in adults with ASD, or whether the apparent decline reflects under ascertainment and diagnostic gaps rather than a genuine reduction in vulnerability (24, 25). In a cross-sectional study of over 9,900 adults aged 50 and over, depression, anxiety, PTSD, loneliness, and social isolation each mediated the relationship between autistic traits and suicidal ideation, while male sex, depression, PTSD, and social isolation specifically mediated suicidal self-harm (26). The vulnerabilities associated with elevated SA risk during the transition to adulthood in ASD — social exclusion, unemployment, and limited social network formation — do not necessarily resolve with age and may instead accumulate across the lifespan in ways that sustain rather than attenuate risk (26, 29, 65). The combination of these ASD-specific vulnerabilities with general population risk factors for suicidality in older adulthood — including bereavement, physical health decline, and reduced help-seeking — may sustain SA risk in older adults with ASD at levels that administrative data systematically underestimate (25, 62). Taken together, these findings highlight the need for lifespan approaches to suicide prevention in ASD that do not treat adolescence and young adulthood as the exclusive window of risk and point to the importance of targeted surveillance and outreach for middle-aged and older adults with ASD — a group that remains dramatically underrepresented in suicide research and prevention efforts (24–26).

4.3 Role of psychiatric comorbidity in suicide attempt prevalence

Having at least one psychiatric comorbidity is associated with a 30-fold higher SA prevalence in our sample (aPR 30.56, 95% CI 26.03–35.89), far exceeding the SA burden observed in individuals with ASD and no comorbidity. This effect is consistent with — though not directly comparable to — general population data showing that any psychiatric disorder is associated with elevated suicide risk, where a 2022 systematic review and meta-analysis found an OR of 13.1 for suicide associated with any psychiatric disorder (27). The larger relative effect observed in our ASD sample likely reflects the particular challenges that psychiatric comorbidity poses in this population — atypical symptom presentation in ASD can complicate the recognition, diagnosis, and treatment of co-occurring conditions, potentially allowing psychiatric illness to go unidentified or undertreated for longer periods and amplifying its impact on SA risk (9, 11, 39). An alternative, non-exclusive explanation is suggested by a study directly contrasting autistic and non-autistic individuals, which found comparable effect magnitudes for psychiatric comorbidity's association with self-harm across groups (28) — raising the possibility that the elevated aPR observed here partly reflects the prevalence of comorbidity within our ASD sample, rather than a comorbidity effect unique to ASD. This is consistent with prior population-based evidence that psychiatric comorbidity is among the most consistently identified correlates of SA in ASD across multiple datasets and diagnostic categories (8, 12, 20). Methodological differences limit direct numeric comparison with general population estimates — our prevalence ratios are calculated within the ASD sample relative to those without comorbidity rather than against a non-ASD referent — and should be interpreted in that context.

SA prevalence varies meaningfully across specific comorbidity types, and the pattern of findings across categories provides clinically important granularity beyond the overall comorbidity effect. Among broad comorbidity categories, substance-related disorders are associated with the highest crude SA prevalence (16.9%; uPR 134.20), and their subtype-level characterization — a novel

contribution of the current study — is discussed in detail in the following section. Schizophrenia and psychotic disorders are associated with the second highest crude SA prevalence (11.8%; uPR 93.89). The elevated SA prevalence in this group is consistent with prior evidence from ASD populations and with the well-established link between psychosis and suicidality in the general population, where hopelessness, depression, and command hallucinations have been identified as key mediating mechanisms (20, 30). Co-occurring psychosis in ASD may represent a particularly high-risk combination, given that ASD-specific vulnerabilities — including discrepancies in how distress is communicated to persons without ASD and limited social support — may compound the suicide risk already elevated by psychosis alone (12, 30). Trauma and stressor-related disorders are associated with an 8.2% crude SA prevalence (uPR 65.22). PTSD has been identified as one of the most strongly associated comorbidities with SA in ASD specifically, reflecting the intersection of trauma exposure with ASD-specific emotion regulation difficulties (11, 20). Mood disorders show a 7.4% crude SA prevalence (uPR 58.95) and represent the most extensively studied comorbidity in the ASD suicidality literature — depression has been identified as one of the most consistently associated correlates of suicidal thoughts and behaviors in ASD across multiple systematic reviews and meta-analyses, and depressive symptoms are thought to interact with ASD-specific cognitive features such as rigidity and perseveration to amplify SA risk (9, 12, 22, 29). Eating disorders show the lowest crude SA prevalence among comorbidity categories (5.0%; uPR 39.78), though they confer markedly elevated SA prevalence relative to those with no comorbidity, and a striking sex difference — females with ASD and eating disorders show a crude SA prevalence of 9.6% compared to 2.0% in males. This sex disparity is consistent with the well-documented phenotypic and etiological overlap between ASD and eating disorders, particularly anorexia nervosa, which affects females at roughly ten times the rate of males (69) and has been associated with elevated SA risk both in the general population and in the context of ASD (12.29).

The association between psychiatric comorbidity and SA prevalence differs by sex — stronger in females with ASD (aPR 36.54) than in males (aPR 27.77), as described in Section 4.1 above. This sex-specific pattern reinforces the broader finding that females with ASD face a compounding burden — higher comorbidity rates, later diagnosis, and a stronger relative impact of comorbidity on SA prevalence — that collectively elevates their SA burden relative to males across the lifespan (12, 20, 21). The age-stratified comorbidity data further show that crude SA prevalence in those with at least one psychiatric comorbidity peaks in the 15–24 age group (7.86% vs. 0.26% without comorbidity) and declines progressively with age, consistent with the overall age pattern and reinforcing the concentration of comorbidity-associated SA burden during adolescence and young adulthood in ASD (9, 22).

4.4 Substance use disorder subtypes and suicide attempt prevalence

Substance-related disorders are associated with the highest SA prevalence among all broad comorbidity categories in our cohort (16.9%; uPR 134.20), consistent with prior evidence that substance use disorders are among the most strongly associated comorbidities with SA in ASD (12, 20) and with the well-established association between substance use disorders and elevated suicide risk in the general population (27, 58). National Medicaid data have documented increasing prevalence of substance use disorder diagnoses among individuals with ASD over time, with cannabis use disorder emerging as the most prevalent subtype in ASD-only populations — a pattern compounded by co-occurring psychiatric disorders including depression (71, 72). The current study extends this prior work by characterizing SA prevalence across specific substance use disorder subtypes in individuals with ASD at population scale — a level of granularity that has not previously been reported in this population and that carries direct relevance for targeted clinical monitoring.

All five substance use disorder subtypes examined are associated with SA prevalence ranging from 19.7% to 24.3%, each far exceeding both the overall substance-related disorder category prevalence of 16.9% and the overall ASD cohort prevalence of 1.7%. This convergence of high SA prevalence across all subtypes is consistent with broader general population evidence that alcohol, cannabis, opioids, and stimulants each independently elevate suicide risk through overlapping but distinct pathways — including substance-induced depression, disinhibition of impulse control, pharmacological effects on mood regulation, and the progressive occupational and social decline that frequently accompanies substance use disorder (27, 58). In individuals with ASD, these general mechanisms may be compounded by pre-existing ASD-specific vulnerabilities — including difficulties with emotion regulation, limited social support networks, and challenges communicating distress — that reduce capacity to buffer the psychological impact of substance use disorder and may amplify the transition from ideation to attempt (10, 11, 20).

Stimulant use disorder is associated with the highest crude SA prevalence of any subtype (24.3%; uPR 196.11) in both females (27.1%) and males (23.0%). In the general population, stimulant use — particularly methamphetamine — is associated with elevated suicidality through direct neurobiological effects on dopaminergic and serotonergic systems that impair mood regulation and impulse control, as well as through stimulant-induced mood disorders including depression and dysphoria during withdrawal (73). In individuals with ASD, pre-existing difficulties with emotional regulation — a characteristic feature of ASD that is further amplified by co-occurring ADHD, the most prevalent ASD comorbidity — may compound stimulant-related dysregulation and lower the threshold for suicidal behavior (5, 11, 22). Cannabis use disorder is associated with high crude SA prevalence in both females (26.1%; uPR 170.70) and males. The association between cannabis use disorder and suicidality is well-documented in the general population, with cannabis use linked to elevated SA prevalence across adolescents and young adults, and females with cannabis use disorder consistently showing higher rates of suicidal ideation and SA than males — a sex difference that is mirrored in our findings and that has been attributed in part to the greater severity of cannabis-associated internalizing symptoms in females, including depression and anxiety (58, 74, 75). In individuals with ASD, cannabis use may carry elevated risk given the elevated rates of cannabis use

disorder documented in this population specifically and the potential for cannabis-induced exacerbation of social withdrawal and mood dysregulation in individuals already vulnerable to these outcomes (11, 71, 72).

Opioid use disorder is associated with a crude SA prevalence of 19.7% (uPR 174.59). One proposed mechanism linking opioid use disorder to suicidality involves social pain — chronic social exclusion is associated with dysfunction in the endogenous opioid system, which in turn contributes to depression and suicidality; opioid use disorder may emerge partly in response to this social pain while simultaneously deepening the underlying vulnerability (76). In individuals with ASD, where chronic social exclusion is among the most consistent features of the condition across the lifespan, this mechanism may carry particular salience and warrants further investigation (29, 65, 76). Alcohol use disorder is associated with the lowest SA prevalence among subtypes in our cohort (19.9%; uPR 158.71), though it remains far above both the overall ASD cohort prevalence and general population estimates. The association between alcohol use disorder and suicidality operates through multiple mechanisms including alcohol-induced depression, cognitive constriction, disinhibition of impulse control during acute intoxication, and acceleration of the transition from suicidal ideation to action — all of which may interact with ASD-specific difficulties in emotion regulation to amplify risk (27, 58).

The consistently higher SA prevalence in females than males across all SUD subtypes is a novel finding that has not previously been characterized in ASD at population scale. Stimulant use disorder shows the highest crude SA prevalence in both females (27.1%) and males (23.0%), while cannabis use disorder shows the second highest in females (26.1%). This sex pattern across all SUD subtypes is consistent with the broader finding throughout our cohort that females with ASD carry a disproportionate SA burden relative to males and likely reflects the compounding of substance use disorder with the ASD-specific vulnerabilities documented in Section 4.1 — including camouflaging demands, later diagnosis, and higher psychiatric comorbidity burden — rather than substance use disorder alone (10, 19, 20). Camouflaging in particular — the suppression or masking of ASD traits to conform to social expectations — has been proposed as a vulnerability that may itself increase risk of substance misuse, in addition to its established association with SA risk, suggesting a complex and potentially bidirectional relationship between these compounding vulnerabilities (77). The age-stratified findings further show that SA prevalence peaks in the 15–24 age group across all SUD subtypes — with stimulant use disorder showing the highest crude SA prevalence in this age group (35.6%), followed by opioid use disorder (31.0%), alcohol use disorder (30.1%), other substances (29.1%), and cannabis use disorder (26.7%). This concentration of SUD-associated SA burden during adolescence and young adulthood is consistent with the overall age pattern described in Section 4.2 and reinforces the importance of substance use disorder screening during this developmental window for individuals with ASD — a period when formal support is declining, psychiatric comorbidities are emerging, and social demands are increasing simultaneously (9, 22, 67). That SA prevalence across SUD subtypes remains elevated across midlife despite progressive decline is consistent with the lifespan vulnerability pattern described in Section 4.2 and underscores that the intersection of substance use disorder and SA risk in ASD is sustained rather than confined to adolescence (20, 24).

4.5 Sensitivity analysis: intellectual disability

Overall SA prevalence was higher among individuals with ID (2.37–2.38%) relative to those without ID (1.58%). At the level of individual comorbidity types, this pattern varied across comorbidity types. Three comorbidities showed a substantial reduction in unadjusted prevalence ratio among individuals with ID relative to those without: eating disorders (uPR 45.23, 95% CI 12.68–161.32, p=4.24e-09 without ID vs. uPR 31.79, 95% CI 14.12–71.55, p<2.20e-16 with ID; 29.7% lower), mood disorders (uPR 61.74, 95% CI 28.94–131.69, p<2.20e-16 without ID vs. uPR 48.25, 95% CI 31.51–73.89, p<2.20e-16 with ID; 21.8% lower), and schizophrenia and psychotic disorders (uPR 103.31, 95% CI 49.46–215.78, p<2.20e-16 without ID vs. uPR 65.93, 95% CI 42.10–103.25, p<2.20e-16 with ID; 36.2% lower). This pattern is consistent with well-documented underdiagnosis of psychiatric disorders in individuals with ID: a diagnostic assessment study identified previously undiagnosed mental disorders in 29.6% of participants with ID (78), attributable in part to diagnostic overshadowing, in which psychiatric symptoms are misattributed to the underlying disability rather than assessed independently (79). Depression in particular has been observed to present atypically in individuals with ID, with irritability, somatic complaints, and behavioral changes such as aggression or withdrawal substituting for the classic affective symptoms used in standard diagnostic criteria (80). If psychiatric comorbidities are systematically underdiagnosed and under coded in individuals with ID, the true comorbidity-associated SA risk in this subgroup would be expected to appear attenuated relative to individuals without ID, consistent with the pattern observed here.

Trauma and stressor-related disorders (uPR 66.20 without ID vs. 66.07 with ID) and substance-related disorders overall (uPR 135.85 without ID vs. 136.13 with ID) showed negligible differences by ID status. Among substance use disorder subtypes, the direction of association varied: alcohol (uPR 158.84 vs. 180.66), cannabis (uPR 170.57 vs. 183.64), and stimulant use disorder (uPR 193.94 vs. 197.58) showed modestly higher uPRs among individuals with ID, whereas opioid use disorder (uPR 170.61 vs. 163.45) and other substances (uPR 186.35 vs. 152.44) showed lower uPRs. We are not aware of an established explanation for this subtype-specific heterogeneity, and interpret it cautiously given the modest magnitude of these differences relative to the psychiatric comorbidity findings above. Interpreting these findings against the existing literature is further complicated by differences in outcome definition: much of the prior work on ID and suicide-related outcomes in ASD examines combined measures of suicidal ideation and self-harm rather than SA specifically (35), making direct comparison to our SA-specific findings difficult, as these outcomes are not interchangeable.

4.6 Sensitivity analysis: race and ethnicity

SA prevalence by race/ethnicity ranged from 0.65–0.92% (Non-Hispanic Other Race) to 2.25–2.28% (Non-Hispanic Multi-race).

Non-Hispanic Multi-race individuals showed the highest prevalence of any category (2.25–2.28%), consistent with prior work finding multiracial adolescents at elevated odds of SA specifically relative to White adolescents in the general population (81). Non-Hispanic White individuals showed the second-highest prevalence (2.22%). National mortality data show suicide death rates have traditionally been higher for non-Hispanic White individuals than for non-Hispanic Black or Hispanic individuals, though death and SA are distinct outcomes not directly comparable (82). Non-Hispanic Black or African American individuals fell in an intermediate range (1.23–1.26%), which may relate to a broader shift in mortality data showing a substantial recent rise in suicide death rates among non-Hispanic Black individuals (82).

Hispanic or Latino individuals showed one of the lowest overall SA prevalence estimates of any category (0.78–0.80%), consistent with a broader pattern in the suicide literature in which Hispanic mortality, including death by suicide, has been found to be lower than would be predicted by socioeconomic disadvantage alone (83). This lower baseline did not extend to comorbidity-associated risk: among Hispanic or Latino individuals specifically, unadjusted prevalence ratios for psychiatric and substance-related comorbidities were among the highest observed of any race/ethnicity category, including the single highest uPR in the entire analysis, for stimulant use disorder (uPR 266.51, 95% CI 120.85–587.73, p<2.20e-16). This pattern differs from prior work reporting lower relative suicide risk among Hispanic individuals conditional on psychiatric comorbidity, relative to non-Hispanic White individuals (83). We are not aware of an established explanation for this divergence and given the smaller sample size and correspondingly wider confidence intervals in this subgroup, this finding should be interpreted as hypothesis-generating rather than conclusive.

Non-Hispanic Asian individuals showed one of the lowest overall SA prevalence estimates (0.70–0.84%). We did not identify literature directly comparable to this finding in the context of ASD and interpret it with the same caution applied to the smaller race/ethnicity categories in this analysis.

Non-Hispanic American Indian/Alaska Native or Native Hawaiian/Pacific Islander (combined) individuals showed a relatively elevated overall prevalence (1.07–1.98%) despite the smallest sample size of any category. This broadly aligns with national mortality data identifying American Indian/Alaska Native individuals as having the highest suicide death rates of any racial/ethnic group in the United States, again noting that death and SA prevalence are separate measures (84, 108). However, comorbidity-type unadjusted prevalence ratios could not be reliably estimated for this group: every comorbidity-type comparison for this population was unreportable due to insufficient cell counts, despite combining these two populations specifically to preserve their representation in this analysis. This outcome illustrates a documented concern in the health equity literature, that small-population groups are frequently absent from reported findings, not by deliberate exclusion, but through the routine application of data suppression and disclosure limitation policies intended to protect individual privacy (47) — a concern made more consequential here given that this is among the populations already known to carry the greatest suicide burden nationally.

4.7 Limitations

This study has several methodological limitations that should be considered when interpreting the findings. The use of aggregated count data precluded individual-level analysis and simultaneous adjustment for multiple psychiatric comorbidities — as individuals may carry more than one psychiatric diagnosis, comorbidity-specific and substance use disorder subtype prevalence ratios are presented unadjusted and reflect the association of each comorbidity type with SA relative to those with no comorbidity, rather than independent effects after mutual adjustment. Age-stratified comorbidity and substance use disorder subtype estimates are similarly unadjusted and should be interpreted as descriptive rather than causal. The cross-sectional design and fixed observation window (2016–2025) prevented calculation of person-time at risk for individual patients, precluding the estimation of true incidence rates and limiting the ability to draw conclusions about temporal relationships between comorbidity onset and SA. This limitation reflects a broader gap in the field: a recent systematic review found that 82.5% of studies examining suicidality in ASD used cross-sectional designs, with longitudinal or cohort studies remaining rare (14).

Individuals with ASD were identified using broad ICD-10-CM diagnostic codes (F84.0, F84.5, F84.8, F84.9) without further stratification by symptom severity, verbal ability, or support needs, resulting in a phenotypically heterogeneous cohort. The methods used to identify individuals with ASD within large administrative and EHR databases vary in sensitivity and specificity depending on the diagnostic strategy used, and studies validating ASD identification in electronic health records have found that relying on ICD-CM codes alone can both include false-positive cases and exclude true cases of ASD, particularly when diagnostic documentation is inconsistent across providers and visit types (85, 86). This heterogeneity likely obscures meaningful within-group variation in SA risk, particularly for minimally verbal individuals with ASD, who have long been described as the 'neglected end of the spectrum' in ASD research and are correspondingly underrepresented in most descriptive and intervention studies (87). This underrepresentation may extend to EHR-based research specifically: communication difficulties are among the most consistently identified barriers to healthcare access and engagement for individuals with intellectual disability and/or ASD, which may result in less frequent or less complete clinical documentation for this subgroup relative to more verbal individuals with ASD (88).

SA were identified using ICD-10-CM codes recorded within the Epic Cosmos system, and this approach is known to undercount true SA prevalence in administrative data — a rapid review of ICD-10-based surveillance found that sensitivity for identifying nonfatal suicidal behaviors using administrative codes is low, with studies reporting sensitivity ranging from 12% to 86% and substantial undercounting of true SA events (51). Attempts occurring outside of participating health systems, prior to the study period, or not resulting in a clinical encounter would not be captured. Additionally, ICD-10-CM codes do not distinguish between suicide attempts and non-suicidal self-injury in some coding contexts, which may introduce misclassification (51). These limitations are

likely to result in underestimation of true SA prevalence across all subgroups examined and may differentially affect older age groups and populations with lower healthcare engagement — including older adults with ASD, who face substantial barriers to diagnosis and clinical contact (25, 26).

A related limitation concerns the timing of psychiatric comorbidity diagnoses relative to ASD diagnosis. It has been suggested that delayed identification of autism in females may result in misdiagnosis with other psychiatric conditions (89); examining this would require comparing the age at which each psychiatric comorbidity was diagnosed relative to ASD diagnosis. However, Epic Cosmos captures diagnostic codes rather than verified clinical diagnoses, and the date a code first appears in a patient's record does not necessarily reflect the true onset or timing of diagnosis — particularly for patients whose care spans multiple institutions or predates inclusion in Cosmos-contributing systems. Given this limitation, we were unable to reliably assess diagnostic timing in the current analysis.

Intellectual disability status was incorporated as a binary indicator in the current analysis rather than stratified by severity, as severity-specific coding is inconsistently captured in administrative and clinical data. A federal report examining intellectual and/or developmental disability (ID/DD) identification in Medicare and Medicaid claims data found substantial variation across studies in which diagnostic codes are used to identify ID/DD, reflecting the broader limitation that ICD codes are designed for billing rather than research purposes (90). Even under rigorous, empiric ascertainment, a Centers for Disease Control and Prevention (CDC) surveillance study using intelligence quotient (IQ)-test-based classification still found severity unspecified for 8% of children with ID due to the absence of a specific IQ score (91) — a rate likely higher in routine clinical coding without systematic testing at every encounter. Population-based evidence suggests that ASD without co-occurring intellectual disability is associated with higher SA risk than ASD with co-occurring intellectual disability, and that this relationship varies by severity of intellectual disability (8, 11) — a distinction our binary classification cannot capture. Cell counts of 10 or fewer were suppressed in accordance with Epic Cosmos data privacy policies and excluded from analyses, which may have introduced some bias in estimates for smaller subgroups, particularly for less common race/ethnicity categories, older age categories and for less prevalent substance use disorder subtypes. The comorbidity-type prevalence ratios reported for the intellectual disability and race/ethnicity sensitivity analyses may be similarly affected: while implausible or non-convergent estimates arising from complete data suppression were excluded, cells with partial suppression may still reflect some degree of undercounting not fully captured by this exclusion criterion. Additionally, because the intellectual disability and race/ethnicity sensitivity analyses were conducted separately, this study cannot assess their joint or intersecting effects on SA prevalence — for example, whether SA risk associated with a given comorbidity differs by race/ethnicity specifically within the subgroup of individuals with ID.

Other investigated risk factors shared between ASD and SA — including advanced paternal and maternal age, prematurity (92), childhood maltreatment (93), family psychiatric history (8, 94), low socioeconomic status (95, 109), prenatal smoking (96), and maternal perinatal depression (97) — were not incorporated into the current analysis. Preliminary assessment indicated that data completeness for these variables in Epic Cosmos was insufficient to support reliable inclusion, and paternal-linkage data in particular are not structurally available in the current dataset (98, 99). These findings should therefore be interpreted with the possibility of residual confounding from these unmeasured factors in mind.

Finally, while Epic Cosmos is one of the largest real-world clinical datasets available — aggregating EHR data from over 2,133 hospitals across the United States — it captures only individuals who have accessed healthcare within participating systems, and factors influencing healthcare engagement can result in certain groups being underrepresented relative to the broader population (100). Individuals with ASD who are undiagnosed, have limited healthcare access, or are seen exclusively outside of Cosmos-affiliated institutions would not be captured, and the cohort may therefore skew toward individuals with greater healthcare utilization and more documented clinical contact. These limitations should inform how findings are generalized to the broader ASD population and highlight the importance of complementing EHR-based analyses with data from community and self-report samples.

5 Conclusion

Prior population-based studies have established that individuals with ASD face elevated SA risk relative to the general population, but the scale, demographic breadth, and clinical granularity needed to characterize how that risk is distributed across subgroups — and to support evidence-based risk stratification — have been largely unavailable (6, 9, 12). The present study addresses this gap using a national EHR cohort of over 2.3 million individuals with ASD, providing simultaneously stratified SA prevalence estimates across sex, age, psychiatric comorbidity type, and substance use disorder subtype that prior studies have not been able to generate at this resolution or scale.

The findings reveal a pattern of variation in SA prevalence across demographic and clinical subgroups that is clinically meaningful and that has important implications for how preventive attention should be directed. Psychiatric comorbidity is associated with a 30-fold higher SA prevalence relative to those without comorbidity (aPR 30.56), with the effect stronger in females (aPR 36.54) than in males (aPR 27.77) — consistent with evidence that females with ASD face a compounding burden through camouflaging, diagnostic delay, and higher comorbidity rates that collectively elevates their SA burden beyond what comorbidity alone would predict (10, 19, 21). Substance-related disorders carry the highest SA prevalence among comorbidity categories, and the subtype-level characterization reported here — with SA prevalence ranging from 19.7% to 24.3% across all five subtypes and stimulant use disorder showing the highest unadjusted prevalence ratio (uPR 196.11) — provides a granularity of clinical information not previously available in ASD (20, 73). SA prevalence peaks in the 15–24

age group overall (aPR 5.14) and differs meaningfully by sex — females peaking earlier at 15–24 years (aPR 4.33) and males later at 25–34 years (aPR 6.08) — a sex-specific divergence in the timing of peak SA prevalence that likely reflects distinct developmental pathways and that has direct implications for how clinical attention should be directed across the lifespan (22, 29, 64). SA prevalence remains elevated well into midlife across all subgroups, and the non-significant association in the 65+ age group overall likely reflects under ascertainment rather than true attenuation of risk — consistent with the well-documented underdiagnosis of ASD in older adults and the known sensitivity limitations of ICD-10-CM coding for nonfatal suicidal behaviors (24, 26, 51).

Supplementary sensitivity analyses examined SA prevalence by intellectual disability (ID) status and race/ethnicity, revealing meaningful, if complex, patterns of variation. SA prevalence was higher among individuals with ID (2.37–2.38%) than without (1.58%), an elevation that did not hold uniformly across comorbidity types: psychiatric comorbidities including mood disorders and schizophrenia showed lower prevalence ratios among individuals with ID, plausibly reflecting the tendency for psychiatric symptoms in this population to be misattributed to the underlying disability rather than independently assessed (78, 79), whereas substance-related comorbidities showed no comparable reduction. Race/ethnicity showed similarly complex patterning: SA prevalence ranged from 0.65–0.92% among Non-Hispanic Other Race individuals to 2.25–2.28% among Non-Hispanic Multi-race individuals (81), and Hispanic or Latino individuals showed a combination of low overall prevalence (83) and disproportionately elevated comorbidity-associated risk that has not, to our knowledge, been previously reported. Comorbidity-type estimates could not be reliably calculated for several smaller race/ethnicity categories, most notably American Indian/Alaska Native and Native Hawaiian/Pacific Islander individuals — a population independently documented to carry among the highest suicide burden nationally (84) — underscoring the need for adequately powered, purpose-built samples to characterize SA risk in historically underrepresented populations.

These findings are directly relevant to suicide risk stratification in ASD — the process of using population-level data on demographic and clinical characteristics to identify which subgroups carry the greatest burden and to direct preventive resources toward them, rather than applying uniform approaches across a heterogeneous population (41, 42). Standard suicide screening tools have not been validated for use in ASD, universal screening has not demonstrated effectiveness in the adult population broadly, and clinicians report lower self-efficacy in assessing suicide risk in individuals with ASD compared to clients without ASD — underscoring the need for stratified, evidence-based approaches that concentrate surveillance and intervention on those at demonstrably elevated risk (36, 39, 40).

Some progress toward addressing this gap is emerging: recently validated ASD-adapted screening instruments, including the Suicidal Behaviours Questionnaire–Autism Spectrum Conditions (101) and the Suicidal Ideation Attributes Scale-Modified (102), along with clinician-reported lower acceptability of standard safety planning interventions for clients with ASD (39), have motivated the development of ASD-adapted safety plans, evaluated in a pilot randomized controlled trial with promising results (40). Sensory and communication needs — including preference for concrete, literal language and visual supports over abstract verbal instruction — have similarly been identified as critical considerations for adapting crisis and safety planning interventions for individuals with ASD (103). Caregiver involvement also emerges as a critical but often overlooked component of care — qualitative work with youth with ASD, caregivers, and clinicians has found that providers frequently fail to incorporate caregiver perspectives into suicide risk assessment and management and has recommended better educating families about ASD-specific suicide risk and addressing caregiver and family stress directly (104). No evidence-based post-discharge intervention has yet been validated specifically for individuals with ASD, though a large ongoing federally-funded trial is currently comparing structured follow-up care against standard safety planning in this population (Patient-Centered Outcomes Research Institute, 2021) — reflecting active progress rather than an unaddressed gap. Suicide prevention research broadly distinguishes interventions by level of targeting — universal strategies applied regardless of individual risk, selective strategies aimed at higher-risk groups and indicated strategies for individuals already showing warning signs (105). No single intervention is likely to be uniformly effective across the heterogeneous ASD population; a recent scoping review of suicide prevention interventions for the autistic community similarly concluded that tailored, individualized approaches are needed rather than a uniform standard (106). Effective prevention will instead require layered, individualized approaches spanning screening, crisis planning, caregiver involvement, and continuity of care.

Complementing these individual-level clinical tools, EHR-based machine learning models have demonstrated strong clinical utility for suicide risk stratification in psychiatric populations, and extending this approach to ASD requires precisely the kind of population-level descriptive foundation that the current study provides (42, 43, 107). The subgroup-level SA prevalence estimates reported here — identifying females with ASD, individuals in the 15–24 and 25–34 age windows, those with co-occurring psychiatric and substance use disorders, individuals with ID, and Non-Hispanic Multi-race individuals, as the groups carrying the greatest SA burden — provide the empirical grounding needed to develop and validate ASD-specific risk stratification tools (11, 45). Future work should extend this foundation in several directions. While intellectual disability status and race and ethnicity were examined here as supplementary sensitivity analyses, fully integrating these variables into the primary risk stratification model, stratifying by intellectual disability severity, and examining their joint or intersecting effects on SA prevalence will be essential to characterizing the full heterogeneity of SA risk within ASD and ensuring that stratification approaches are equitable across subpopulations (8, 62). Longitudinal analyses examining trajectories of SA risk over time, the temporal relationship between comorbidity onset and SA, and the impact of ASD-specific clinical interventions on SA outcomes would meaningfully complement the cross-sectional prevalence estimates reported here (9, 11, 40). Community and self-report samples are also needed to reach individuals who do not

access healthcare within EHR-affiliated systems — particularly older adults with ASD, who face considerable barriers to diagnosis and clinical engagement and whose SA burden is likely underestimated in administrative data (25, 26). Taken together, the present findings establish a population-level empirical foundation for more targeted, evidence-based, and clinically meaningful approaches to suicide prevention in ASD — a population carrying a significant and preventable burden of suicidal behavior that has been insufficiently characterized and insufficiently addressed.

Data availability statement

The datasets presented in this study are derived from the Epic Cosmos data warehouse and are not publicly available due to data privacy and security restrictions inherent to the Cosmos Data Science Virtual Machine (DSVM) environment. Researchers interested in accessing Epic Cosmos data should contact Epic Systems (https://cosmos.epic.com) or their participating institution. The analytic code used in this study is available from the de Lacy lab's GitHub (delacylab/ASD_Suicide_Prevalence_EHR_Data). Further inquiries can be directed to the corresponding author.

Ethics statement

The requirement of ethical approval was waived by University of Utah Institutional Review Board for the studies involving humans because This study used de identified data from the EPIC Cosmos Database. The studies were conducted in accordance with the local legislation and institutional requirements. The ethics committee/institutional review board also waived the requirement of written informed consent for participation from the participants or the participants' legal guardians/next of kin because This study was determined to not meet the definitions of Human Subjects Research by the University of Utah Institutional Review Board (IRB_00202177) and was therefore exempt from informed consent requirements.

Author contributions

MB: Visualization, Formal Analysis, Validation, Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Investigation, Software. W-yL: Writing – review & editing, Investigation, Resources, Data curation, Methodology. MV: Software, Data curation, Writing – review & editing, Formal Analysis. NL: Conceptualization, Funding acquisition, Writing – review & editing, Project administration, Validation, Methodology, Supervision.

Funding

The author(s) declared that financial support was received for this work and/or its publication. The author(s) received financial support from the Huntsman Mental Health Foundation.

Acknowledgments

The authors would like to thank David Danks for his contributions to the content of the methodology and the manuscript.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. The authors acknowledge the use of Claude (Anthropic, San Franciso, CA) for assistance with language editing and manuscript revision. All content was reviewed, verified, and approved by the authors, who take responsibility for the accuracy of the published work.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fpsyt.2026.1940193/full#supplementary-material

References

1. Hodges H, Fealko C, Soares N. Autism spectrum disorder: definition, epidemiology, causes, and clinical evaluation. *Transl Pediatr*. (2020) 9(Suppl 1):S55–65. doi: 10.21037/tp.2019.09.09

2. Masi A, DeMayo MM, Glozier N, Guastella AJ. An overview of autism spectrum disorder, heterogeneity and treatment options. *Neurosci Bull*. (2017) 33:183–93. doi: 10.1007/s12264-017-0100-y

3. American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders (5th Ed. Text Rev.)*. Washington, DC: American Psychiatric Association Publishing (2022). doi: 10.1176/appi.books.9780890425787

4. Rylaarsdam L, Guemez-Gamboa A. Genetic causes and modifiers of autism spectrum disorder. *Front Cell Neurosci*. (2019) 13:385. doi: 10.3389/fncel.2019.00385

5. Khachadourian V, Mahjani B, Sandin S, Kolevzon A, Buxbaum JD, Reichenberg A, et al. Comorbidities in autism spectrum disorder and their etiologies. *Transl Psychiatry*. (2023) 13:71. doi: 10.1038/s41398-023-02374-w

6. Huntjens A, Landlust A, Wissenburg S, van der Gaag M. The prevalence of suicidal behavior in autism spectrum disorder: a meta-analysis. *Crisis*. (2024) 45:144–53. doi: 10.1027/0227-5910/a000922

7. Santomauro DF, Hedley D, Sahin E, Brugha TS, Naghavi M, Vos T, et al. The global burden of suicide mortality among people on the autism spectrum: a systematic review, meta-analysis, and extension of estimates from the Global Burden of Disease Study 2021. *Psychiatry Res*. (2024) 341:116150. doi: 10.1016/j.psychres.2024.116150

8. Hirvikoski T, Boman M, Chen Q, D'Onofrio BM, Mittendorfer-Rutz E, Lichtenstein P, et al. Individual risk and familial liability for suicide attempt and suicide in autism: a population-based study. *Psychol Med*. (2020) 50:1463–74. doi: 10.1017/s0033291719001405

9. Brown CM, Newell V, Sahin E, Hedley D. Updated systematic review of suicide in autism: 2018–2024. *Curr Dev Disord Rep*. (2024) 11:225–56. doi: 10.1007/s40474-024-00308-9

10. Cassidy S, McLaughlin E, McGranaghan R, Pelton M, O'Connor R, Rodgers J. Is camouflaging autistic traits associated with defeat, entrapment, and lifetime suicidal thoughts? Expanding the Integrated Motivational Volitional Model of Suicide. *Suicide Life Threat Behav*. (2023) 53:572–85. doi: 10.1111/sltb.12965

11. Reid M, Delgado D, Heinly J, Kiernan B, Shapiro S, Morgan L, et al. Suicidal thoughts and behaviors in people on the autism spectrum. *Curr Psychiatry Rep*. (2024) 26:563–72. doi: 10.1007/s11920-024-01533-0

12. Kim JH, Lee J, Shim S, Cheon KA. Association of self-harm and suicidality with psychiatric co-occurring conditions in autistic individuals: a systematic review and pooled analysis. *eClinicalMedicine*. (2024) 77:102863. doi: 10.1016/j.eclinm.2024.102863

13. Lai MC, Saunders NR, Huang A, Artani A, Wilton AS, Zaheer J, et al. Self-harm events and suicide deaths among autistic individuals in Ontario, Canada. *JAMA Netw Open*. (2023) 6:e2327415. doi: 10.1001/jamanetworkopen.2023.27415

14. Newell V, Phillips L, Jones C, Townsend E, Richards C, Cassidy S. A systematic review and meta-analysis of suicidality in autistic and possibly autistic people without co-occurring intellectual disability. *Mol Autism*. (2023) 14:12. doi: 10.1186/s13229-023-00544-7

15. Etherson ME, Lee S, Loney KJ, Steward IP, Ward J, McClelland H, et al. Exploring risk and protective factors which distinguish suicidal and self-harm behaviours from suicidal and self-harm ideation in young people: a systematic review. *PloS One*. (2025) 20:e0326381. doi: 10.1371/journal.pone.0326381

16. Pollak OH, Sheehan AE, Walsh RFL, Stephenson AR, Zell H, Mayes J, et al. Assessment of suicidal thoughts and behaviors in adults: a systematic review of measure psychometric properties and implications for clinical and research utility. *Clin Psychol Rev*. (2024) 112:102464. doi: 10.1016/j.cpr.2024.102464

17. Nobile B, Gourguechon-Buot E, Olié E, Courtet P. Risk factors of transition from suicidal ideation to suicide attempt: a one-year longitudinal study among hospital-based in- and outpatients. *J Affect Disord*. (2025) 390:119854. doi: 10.1016/j.jad.2025.119854

18. Kirby AV, Conner CM, Mazefsky CA. Are autistic females at greater risk of suicide? A call for clarity to advance suicide prevention for the whole community. *Autism Res*. (2024) 17:898–905. doi: 10.1002/aur.3120

19. Hull L, Stark I, Lundberg M, Ahlqvist VH, Idring Nordström S, Ohlis A, et al. Sex differences in self-harm and suicide in young autistic adults. *Acta Psychiatr Scand*. (2024) 150(4):223–33. doi: 10.1111/acps.13736

20. Kõlves K, Fitzgerald C, Nordentoft M, Wood SJ, Erlangsen A. Assessment of suicidal behaviors among individuals with autism spectrum disorder in Denmark. *JAMA Netw Open*. (2021) 4(1):e2033565. doi: 10.1001/jamanetworkopen.2020.33565

21. Harrop C, Tomaszewski B, Putnam O, Klein C, Lamarche E, Klinger L. Are the diagnostic rates of autistic females increasing? An examination of state-wide trends. *J Child Psychol Psychiatry*. (2024) 65:973–83. doi: 10.1111/jcpp.13939

22. Cook ML, Tomaszewski B, Lamarche E, Bowman K, Klein CB, Stahl S, et al. Suicide risk in transition-aged autistic youth: the link among executive function, depression, and autistic traits. *Autism*. (2024) 28:2311–21. doi: 10.1177/13623613241227983

23. Chen MH, Pan TL, Lan WH, Hsu JW, Huang KL, Su TP, et al. Risk of suicide attempts among adolescents and young adults with autism spectrum disorder: a nationwide longitudinal follow-up study. *J Clin Psychiatry*. (2017) 78(9):e1174-9. doi: 10.4088/JCP.16m11100

24. Mason D, Stewart GR, Capp SJ, Happé F. Older age autism research: a rapidly growing field, but still a long way to go. *Autism Adulthood*. (2022) 4:164–72. doi: 10.1089/aut.2021.0041

25. Roper S, Charlton RA, Happé F, Stewart GR. Self-harm and suicidality experiences of autistic and non-autistic adults in midlife and old age. *Mol Autism*. (2025) 16:58. doi: 10.1186/s13229-025-00693-x

26. Nuzum E, Medeisyte R, Eshetu A, Hoare S, Corbett A, Ballard C, et al. Autistic traits and suicidality in midlife and old age: investigating mediating effects of mental health and social connectedness. *Nat Ment Health*. (2026) 4(2):255–62. doi: 10.1038/s44220-025-00579-0

27. Favril L, Yu R, Uyar A, Sharpe M, Fazel S. Risk factors for suicide in adults: systematic review and meta-analysis of psychological autopsy studies. *BMJ Ment Health*. (2022) 25(4):148–55. doi: 10.1136/ebmental-2022-300549

28. Jokiranta-Olkoniemi E, Gyllenberg D, Sucksdorff D, Suominen A, Kronström K, Chudal R, et al. Risk for premature mortality and intentional self-harm in autism spectrum disorders. *J Autism Dev Disord*. (2021) 51:3098–108. doi: 10.1007/s10803-020-04768-x

29. Pelton MK, Crawford H, Robertson AE, Rodgers J, Baron-Cohen S, Cassidy S. Understanding suicide risk in autistic adults: comparing the interpersonal theory of suicide in autistic and non-autistic samples. *J Autism Dev Disord*. (2020) 50(10):3620–37. doi: 10.1007/s10803-020-04393-8

30. Upthegrove R, Abu-Akel A, Chisholm K, Lin A, Zahid S, Pelton M, et al. Autism and psychosis: clinical implications for depression and suicide. *Schizophr Res*. (2018) 195:80–5. doi: 10.1016/j.schres.2017.08.028

31. Foye U, Kakar S, McNamara N, Musial A, Jewell T, Griffiths JR, et al. 'It's the perfect storm': why are people with eating disorders at risk of suicide? A qualitative study. *BMC Med*. (2025) 23:481. doi: 10.1186/s12916-025-04326-1

32. Ivey-Stephenson AZ, Demissie Z, Crosby AE, Stone DM, Gaylor E, Wilkins N, Lowry R, Brown M. Suicidal ideation and behaviors among high school students — Youth Risk Behavior Survey, United States, 2019. *MMWR Suppl*. (2020) 69(1):47–55. doi: 10.15585/mmwr.su6901a6

33. Williams EG, Sivathasan S, Anthony N, Eack SM, Mazefsky CA. Racial disparities in depression and mental health service use among Black and White autistic adults. *Sci Rep*. (2026) 16:4617. doi: 10.1038/s41598-025-34696-8

34. Hand BN, Benevides TW, Carretta HJ. Suicidal ideation and self-inflicted injury in Medicare enrolled autistic adults with and without co-occurring intellectual disability. *J Autism Dev Disord*. (2020) 50:3489–95. doi: 10.1007/s10803-019-04345-x

35. Cervantes PE, Brown DS, Horwitz SM. Suicidal ideation and intentional self-inflicted injury in autism spectrum disorder and intellectual disability: an examination of trends in youth emergency department visits in the United States from 2006 to 2014. *Autism*. (2023) 27:226–43. doi: 10.1177/13623613221091316

36. US Preventive Services Task Force, Barry MJ, Nicholson WK, Silverstein M, Chelmow D, Coker TR, et al. Screening for depression and suicide risk in adults: US Preventive Services Task Force recommendation statement. *JAMA*. (2023) 329(23):2057–67. doi: 10.1001/jama.2023.9297

37. O'Connor E, Henninger M, Perdue LP, Coppola EL, Durbin S, Thomas R, et al. *Screening for Depression, Anxiety, and Suicide Risk in Adults: A Systematic Evidence Review for the U.S. Preventive Services Task Force. Evidence Synthesis No. 223*. Rockville, MD: Agency for Healthcare Research and Quality (US) (2023). AHRQ publication 22-05295-EF-1.

38. Vasa RA, Kalari VK, Kitchen CA, Kharrazi H, Campo JV, Wilcox HC. Suicide risk screening in children and adolescents with autism spectrum disorder presenting to the emergency department. *Jt Comm J Qual Patient Saf*. (2025) 51(3):192–8. doi: 10.1016/j.jcjq.2024.11.002

39. Jager-Hyman S, Maddox BB, Crabbe SR, Mandell DS. Mental health clinicians' screening and intervention practices to reduce suicide risk in autistic adolescents and adults. *J Autism Dev Disord*. (2020) 50:3450–61. doi: 10.1007/s10803-020-04441-3

40. Rodgers J, Cassidy S, Pelton M, Goodwin J, Wagnild J, Bhattarai N, et al. Feasibility and acceptability of autism adapted safety plans: an external pilot randomised controlled trial. *eClinicalMedicine*. (2024) 73:102662. doi: 10.1016/j.eclinm.2024.102662

41. Castagna C, Huff A, Douglas A, Garofano M, Fabi M, Hass R, et al. Stratifying the population based on health risk: identification of patient key health risk factors through consensus techniques. *BMC Prim Care*. (2025) 26:229. doi: 10.1186/s12875-025-02923-w

42. Tran T, Luo W, Phung D, Harvey R, Berk M, Kennedy RL, et al. Risk stratification using data from electronic medical records better predicts suicide risks than clinician assessments. *BMC Psychiatry*. (2014) 14:76. doi: 10.1186/1471-244X-14-76

43. Barak-Corren Y, Castro VM, Nock MK, Mandl KD, Madsen EM, Seiger A, et al. Validation of an electronic health record-based suicide risk prediction modeling approach across multiple health care systems. *JAMA Netw Open*. (2020) 3(3):e201262. doi: 10.1001/jamanetworkopen.2020.1262

44. McCarthy JF, Cooper SA, Dent KR, Eagan AE, Matarazzo BB, Hannemann CM, et al. Evaluation of the Recovery Engagement and Coordination for Health–Veterans

Enhanced Treatment suicide risk modeling clinical program in the Veterans Health Administration. *JAMA Netw Open*. (2021) 4(10):e2129900. doi: 10.1001/jamanetworkopen.2021.29900

45. Brown KA, Donise KR, Cancilliere MK, Aluthge DP, Chen ES. Characterizing autism spectrum disorder and predicting suicide risk for pediatric psychiatric emergency services encounters. *AMIA Annu Symp Proc*. (2024) 2023:864–73.

46. Epic Systems Corporation. *Cosmos: Real-World Data Powered by the Healthcare Community* (2024). Available online at: https://cosmos.epic.com (Accessed July 15, 2026).

47. Mays VM, Echo-Hawk A, Cochran SD, Akee R. Data equity in American Indian/Alaska Native populations: respecting Sovereign Nations' right to meaningful and usable COVID-19 data. *Am J Public Health*. (2022) 112:1416–20. doi: 10.2105/AJPH.2022.307043

48. Dunn DS, Andrews EE. Person-first and identity-first language: developing psychologists' cultural competence using disability language. *Am Psychol*. (2015) 70:255–64. doi: 10.1037/a0038636

49. Taboas A, Doepke K, Zimmerman C. Preferences for identity-first versus person-first language in a US sample of autism stakeholders. *Autism*. (2023) 27:565–70. doi: 10.1177/13623613221130845

50. Hedegaard H, Curtin SC, Warner M. *Suicide mortality in the United States, 1999 -2017. NCHS Data Brief, no 330*. Hyattsville, MD: National Center for Health Statistics (2018).

51. Hensley SW, Esie P, Ta M, Matheson A. Accuracy of suicidal behaviors in administrative data as measured by International Classification of Diseases, Tenth Revision–based codes, 2000–2024: a rapid review. *Public Health Rep*. (2025) 2025:00333549251350843. doi: 10.1177/00333549251350843

52. Wilson EB. Probable inference, the law of succession, and statistical inference. *J Am Stat Assoc*. (1927) 22:209–12. doi: 10.2307/2276774

53. Zou G. A modified Poisson regression approach to prospective studies with binary data. *Am J Epidemiol*. (2004) 159:702–6. doi: 10.1093/aje/kwh090

54. Barros AJD, Hirakata VN. Alternatives for logistic regression in cross-sectional studies: an empirical comparison of models that directly estimate the prevalence ratio. *BMC Med Res Methodol*. (2003) 3:21. doi: 10.1186/1471-2288-3-21

55. Seabold S, Perktold J. (2010). "Statsmodels: econometric and statistical modeling with Python". In: van der Walt S, Millman J, editors. *Proceedings of the 9th Python in Science Conference*. Austin, TX (2010). p. 92–96. doi: 10.25080/Majora-92bf1922-011

56. Harris-Kojetin BA, Alvey WL, Carlson L, Cohen SB, Cohen SH, Cox LH, et al. *Statistical Policy Working Paper 22: Report on Statistical Disclosure Limitation Methodology*. Washington, DC: Federal Committee on Statistical Methodology, U.S. Office of Management and Budget (2005).

57. Substance Abuse and Mental Health Services Administration. *Key Substance Use and Mental Health Indicators in the United States: Results From the 2024 National Survey on Drug Use and Health*. Rockville, MD: Center for Behavioral Health Statistics and Quality, SAMHSA (2025). Available online at: https://www.samhsa.gov/data/data-we-collect/nsduh-national-survey-drug-use-and-health/national-releases/2024 (Accessed July 15, 2026).

58. Samples H, Cruz N, Corr A, Akkas F. National Trends and Disparities in Suicidal Ideation, Attempts, and Health Care Utilization Among U.S. Adults. *Psychiatr Serv*. (2025) 76(2):110–19. doi: 10.1176/appi.ps.20230466

59. Croen LA, Zerbo O, Qian Y, Massolo ML, Rich S, Sidney S, et al. The health status of adults on the autism spectrum. *Autism*. (2015) 19:814–23. doi: 10.1177/1362361315577517

60. Lai MC, Baron-Cohen S, Buxbaum JD. Understanding autism in the light of sex/gender. *Mol Autism*. (2015) 6:24. doi: 10.1186/s13229-015-0021-4

61. Loomes R, Hull L, Mandy WPL. What is the male-to-female ratio in autism spectrum disorder? A systematic review and meta-analysis. *J Am Acad Child Adolesc Psychiatry*. (2017) 56:466–74. doi: 10.1016/j.jaac.2017.03.013

62. Garnett MF, Curtin SC. Suicide mortality in the United States, 2002–2022. *NCHS Data Brief, no 509*. Hyattsville, MD: National Center for Health Statistics (2024). doi: 10.15620/cdc/160504.

63. Bargiela S, Steward R, Mandy W. The experiences of late-diagnosed women with autism spectrum conditions: an investigation of the female autism phenotype. *J Autism Dev Disord*. (2016) 46:3281–94. doi: 10.1007/s10803-016-2872-8

64. Milner V, Colvert E, Hull L, Cook J, Ali D, Mandy W, et al. Does camouflaging predict age at autism diagnosis? A comparison of autistic men and women. *Autism Res*. (2024) 17:371–81. doi: 10.1002/aur.3059

65. Moseley RL, Gregory NJ, Smith P, Allison C, Cassidy S, Baron-Cohen S. The relevance of the interpersonal theory of suicide for predicting past-year and lifetime suicidality in autistic adults. *Mol Autism*. (2022) 13:14. doi: 10.1186/s13229-022-00495-5

66. Tse VWS, Lei J, Crabtree J, Mandy W, Stott J. Characteristics of older autistic adults: a systematic review of literature. *Rev J Autism Dev Disord*. (2022) 9:184–207. doi: 10.1007/s40489-021-00238-x

67. Laxman DJ, Taylor JL, DaWalt LS, Greenberg JS, Mailick MR. Loss in services precedes high school exit for teens with autism spectrum disorder: a longitudinal study. *Autism Res*. (2019) 12:911–21. doi: 10.1002/aur.2113

68. Kirby AV, Bakian AV, Zhang Y, Bilder DA, Keeshin BR, Coon H. A 20-year study of suicide death in a statewide autism population. *Autism Res*. (2019) 12:658–66. doi: 10.1002/aur.2076

69. Keski-Rahkonen A, Mustelin L. Epidemiology of eating disorders in Europe: prevalence, incidence, comorbidity, course, consequences, and risk factors. *Curr Opin Psychiatry*. (2016) 29:340–5. doi: 10.1097/YCO.0000000000000278

70. Maïano C, Normand CL, Salvas MC, Moullec G, Aimé A. Prevalence of school bullying among youth with autism spectrum disorders: a systematic review and meta-analysis. *Autism Res*. (2016) 9:601–15. doi: 10.1002/aur.1568

71. Roux AM, Tao S, Marcus S, Lushin V, Shea LL. A national profile of substance use disorder among Medicaid enrollees on the autism spectrum or with intellectual disability. *Disabil Health J*. (2022) 15:101289. doi: 10.1016/j.dhjo.2022.101289

72. Lushin V, Marcus S, Tao S, Engstrom M, Roux A, Shea L. Comparing the prevalence of substance use disorders between persons with and without autism spectrum disorders. *Autism*. (2025) 29(7):1674–87. doi: 10.1177/13623613251325282

73. Aljadani AH, Abouzed M, Adam H, Alrasheedy BB, Alarfaj M, Alshammari SR, et al. Methamphetamine use and suicide risk: a comprehensive case-control study. *Front Public Health*. (2025) 13:1593935. doi: 10.3389/fpubh.2025.1593935

74. Han B, Compton WM, Einstein EB, Volkow ND. Associations of Suicidality Trends With Cannabis Use as a Function of Sex and Depression Status. *JAMA Netw Open*. (2021) 4(6):e2113025. doi: 10.1001/jamanetworkopen.2021.13025

75. Oladunjoye AF, Li E, Aneni K, Onigu-Otite E. Cannabis use disorder, suicide attempts, and self-harm among adolescents: a national inpatient study across the United States. *PloS One*. (2023) 18:e0292922. doi: 10.1371/journal.pone.0292922

76. Christie NC. The role of social isolation in opioid addiction. *Soc Cognit Affect Neurosci*. (2021) 16:645–56. doi: 10.1093/scan/nsab029

77. Nel J, Spedding M, Malcolm-Smith S. Consolidating a framework of autistic camouflaging strategies: an integrative systematic review. *Autism*. (2025) 29:2379–94. doi: 10.1177/13623613251335472

78. Peña-Salazar C, Arrufat F, Santos JM, Fontanet A, González-Castro G, Más S, et al. Underdiagnosis of psychiatric disorders in people with intellectual disabilities: differences between psychiatric disorders and challenging behaviour. *J Intellect Disabil*. (2020) 24:326–38. doi: 10.1177/1744629518798259

79. Reiss S, Levitan GW, Szyszko J. Emotional disturbance and mental retardation: diagnostic overshadowing. *Am J Ment Defic*. (1982) 86:567–74.

80. McBrien JA. Assessment and diagnosis of depression in people with intellectual disability. *J Intellect Disabil Res*. (2003) 47:1–13. doi: 10.1046/j.1365-2788.2003.00455.x

81. Cruz N, Duberstein P, Samples H. Racial/ethnic differences in risk of suicidal ideation and suicide attempts in a nationally representative sample of U.S. adolescents. *J Affect Disord*. (2024) 365:32–5. doi: 10.1016/j.jad.2024.08.034

82. Curtin SC, Brown KA, Jordan ME. *Suicide rates for the three leading methods by race and ethnicity: United States, 2000 -2020. NCHS Data Brief, no 450*. Hyattsville, MD: National Center for Health Statistics (2022). doi: 10.15620/cdc:121798

83. Silva C, Van Orden KA. Suicide among hispanics in the United States. *Curr Opin Psychol*. (2018) 22:44–9. doi: 10.1016/j.copsyc.2017.07.013

84. Leavitt RA, Ertl A, Sheats K, Petrosky E, Ivey-Stephenson A, Fowler KA. *Suicides among American Indian/Alaska Natives – National Violent Death Reporting System, 18 states, 2003 -2014. MMWR Morb Mortal Wkly Rep*. (2018) 67:237–42. doi: 10.15585/mmwr.mm6708a1

85. Grosse SD, Nichols P, Nyarko K, Maenner M, Danielson ML, Shea L. Heterogeneity in Autism Spectrum Disorder Case-Finding Algorithms in United States Health Administrative Database Analyses. *J Autism Dev Disord*. (2022) 52:4150–63. doi: 10.1007/s10803-021-05269-1

86. Malow BA, Veatch OJ, Niu X, Fitzpatrick KA, Hucks D, Maxwell-Horn A, et al. A practical approach to identifying autistic adults within the electronic health record. *Autism Res*. (2023) 16:52–65. doi: 10.1002/aur.2849

87. Tager-Flusberg H, Kasari C. Minimally verbal school-aged children with autism spectrum disorder: the neglected end of the spectrum. *Autism Res*. (2013) 6:468–78. doi: 10.1002/aur.1329

88. Doherty M, Neilson S, O'Sullivan J, Carravallah L, Johnson M, Cullen W, et al. Barriers to healthcare and self-reported adverse outcomes for autistic adults: a cross-sectional study. *BMJ Open*. (2022) 12:e056904. doi: 10.1136/bmjopen-2021-056904

89. Kentrou V, Livingston LA, Grove R, Hoekstra RA, Begeer S. Perceived misdiagnosis of psychiatric conditions in autistic adults. *eClinicalMedicine*. (2024) 77:102586. doi: 10.1016/j.eclinm.2024.102586

90. Sommers A, Chiri G, Rios N, Hayes M, Plourde E, Karimi M, et al. *Operationalizing the definition of intellectual and developmental disabilities in administrative claims data for research. ASPE Issue Brief*. Washington, DC: US DHHS Office of the Assistant Secretary for Planning and Evaluation (2025). Available online at: https://aspe.hhs.gov/reports/definition-iddd-administrative-claims-data (Accessed July 15, 2026).

91. Patrick ME, Shaw KA, Dietz PM, Baio J, Yeargin-Allsopp M, Bilder DA, et al. Prevalence of intellectual disability among eight-year-old children from selected communities in the United States, 2014. *Disabil Health J*. (2021) 14:101023. doi: 10.1016/j.dhjo.2020.101023

92. Kolevzon A, Gross R, Reichenberg A. Prenatal and perinatal risk factors for autism: a review and integration of findings. *Arch Pediatr Adolesc Med*. (2007) 161:326–33. doi: 10.1001/archpedi.161.4.326

93. Liu J, Fang Y, Gong J, Cui X, Meng T, Xiao B, et al. Associations between suicidal behavior and childhood abuse and neglect: a meta-analysis. *J Affect Disord*. (2017) 220:147–55. doi: 10.1016/j.jad.2017.03.060

94. Mullins N, Forstner AJ, O'Connell KS, Coombes B, Coleman JRI, Qiao Z, et al. Genome-wide association study of more than 40,000 bipolar disorder cases provides new insights into the underlying biology. *Nat Genet*. (2021) 53:817–29. doi: 10.1038/s41588-021-00857-4

95. Larsson H, Sariaslan A, Långström N, D'Onofrio B, Lichtenstein P. Family income in early childhood and subsequent attention deficit/hyperactivity disorder: a quasi-experimental study. *J Child Psychol Psychiatry*. (2014) 55:428–35. doi: 10.1111/jcpp.12140

96. Rosen BN, Lee BK, Lee NL, Yang Y, Burstyn I. Maternal smoking and autism spectrum disorder: a meta-analysis. *J Autism Dev Disord*. (2015) 45:1689–98. doi: 10.1007/s10803-014-2327-z

97. Tusa BS, Alati R, Betts K, Ayano G, Dachew B. Associations of maternal perinatal depressive disorders with autism spectrum disorder in offspring: findings from a data-linkage cohort study. *Aust N Z J Psychiatry*. (2025) 59:282–92. doi: 10.1177/00048674251315641

98. Kukhareva PV, O'Brien MJ, Malone DC, Kawamoto K, Gouripeddi R, Reddy D, et al. Documentation of social determinants of health for patients with type 2 diabetes in Epic Cosmos. *JAMIA Open*. (2025) 8:ooaf095. doi: 10.1093/jamiaopen/ooaf095

99. Leonard SA, Dysart K, Callahan A, Siadat S, Zhang J, Handley SC, et al. Development of longitudinal, linked maternal-infant cohorts using the Epic Cosmos electronic health record dataset. *medRxiv [Preprint]*. (2026). doi: 10.64898/2026.06.02.26354757

100. Tarabichi Y, Frees A, Honeywell S, Huang C, Naidech AM, Moore JH, et al. The Cosmos collaborative: a vendor-facilitated electronic health record data aggregation platform. *ACI Open*. (2021) 5:e36–46. doi: 10.1055/s-0041-1731004

101. Cassidy SA, Bradley L, Cogger-Ward H, Rodgers J. Development and validation of the Suicidal Behaviours Questionnaire -Autism Spectrum Conditions (SBQ-ASC) in a community sample of autistic, possibly autistic, and non-autistic adults. *Mol Autism*. (2021) 12:46. doi: 10.1186/s13229-021-00449-3

102. Hedley D, Batterham PJ, Bury SM, Clapperton A, Denney K, Dissanayake C, et al. The Suicidal Ideation Attributes Scale-Modified (SIDAS-M): Development and preliminary validation of a new scale for the measurement of suicidal ideation in autistic adults. *Autism*. (2023) 27:1115–31. doi: 10.1177/13623613221131234

103. Schwartzman JM, Smith JR, Bettis AH. Safety planning for suicidality in autism: obstacles, potential solutions, and future directions. *Pediatrics*. (2021) 148:e2021052958. doi: 10.1542/peds.2021-052958

104. Cervantes PE, Palinkas LA, Conlon GR, Richards-Rachlin S, Sullivan KA, Baroni A, et al. Improving emergency department care for suicidality in autism: perspectives from autistic youth, caregivers, and clinicians. *J Autism Dev Disord*. (2025) 55:2820–33. doi: 10.1007/s10803-024-06364-9

105. Sakashita T, Oyama H. Developing a hypothetical model for suicide progression in older adults with universal, selective, and indicated prevention strategies. *Front Psychiatry*. (2019) 10:161. doi: 10.3389/fpsyt.2019.00161

106. Russell A, Rainbow E, Cremen C, Melia R. Suicide Prevention Interventions and Supports for the Autistic Community: A Scoping Review. *Rev J Autism Dev Disord*. (2025). doi: 10.1007/s40489-025-00535-9

107. de Lacy N, Lam WY, Virtosu M, Deshmukh V, Wilson FA, Pescosolido B, et al. Precision stratification of risk for suicidal behavior in people with bipolar depression. In: *Medrxiv* (2026). doi: 10.64898/2026.02.23.26346921

108. Stone D, Trinh E, Zhou H, Welder L, End of Horn P, Fowler K, et al. Suicides among American Indian or Alaska Native persons — National Violent Death Reporting System, United States, 2015–2020. *MMWR Morb Mortal Wkly Rep*. (2022) 71:1161–8. doi: 10.15585/mmwr.mm7137a1

109. Mayes SD, Gorman AA, Hillwig-Garcia J, Syed E. Suicide ideation and attempts in children with autism. *Res Autism Spectr Disord*. (2013) 7:109–19. doi: 10.1016/j.rasd.2012.07.009